

Spring 2026 NB Learning objectives

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0542
SOM.MK.I.BPM2.1.NB.1.NSCI.0543
SOM.MK.I.BPM2.1.NB.1.NSCI.0544
SOM.MK.I.BPM2.1.NB.1.NSCI.0545
SOM.MK.I.BPM2.1.NB.1.NSCI.0546
SOM.MK.I.BPM2.1.NB.4.NSCI.0547

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0001
SOM.MK.I.BPM2.1.NB.1.NSCI.0550
SOM.MK.I.BPM2.1.NB.1.NSCI.0008
SOM.MK.I.BPM2.1.NB.1.NSCI.0009
SOM.MK.I.BPM2.1.NB.1.NSCI.0021
SOM.MK.I.BPM2.1.NB.1.NSCI.0023
SOM.MK.I.BPM2.1.NB.1.NSCI.0024
SOM.MK.I.BPM2.1.NB.1.NSCI.0551

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0022
SOM.MK.I.BPM2.1.NB.1.NSCI.0005
SOM.MK.I.BPM2.1.NB.1.NSCI.0552
SOM.MK.I.BPM2.1.NB.1.NSCI.0004
SOM.MK.I.BPM2.1.NB.1.NSCI.0006
SOM.MK.I.BPM2.1.NB.1.NSCI.0553
SOM.MK.I.BPM2.1.NB.1.NSCI.0554

DLA NSCI
SOM.CS.I.BPM2.1.NB.1.NSCI.0048
SOM.CS.I.BPM2.1.NB.1.NSCI.0049
SOM.CS.I.BPM2.1.NB.1.NSCI.0050

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0070
SOM.MK.I.BPM2.2.NB.1.NSCI.0071
SOM.MK.I.BPM2.1.NB.1.NSCI.0072
SOM.MK.I.BPM2.1.NB.1.NSCI.0562
SOM.MK.I.BPM2.1.NB.1.NSCI.0073
SOM.MK.I.BPM2.1.NB.1.NSCI.0074

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0476

Neuroembryology

Describe the processes underlying neurulation
Describe the formation of the vessicles of the brain
Describe the morphogenesis of the forebrain
Describe the morphogenesis of the brainstem
Describe the morphogenesis of the spinal cord
Describe structural abnormalities in the developing nervous system

General Morphology of the Nervous System

Describe the divisions of the nervous system
Describe the ventricular system of the central nervous system and the relationship of its elements to the embryological origin of the five parts of the brain
Apply the use of anatomical vectors in the representation of the nervous system
Apply the use of anatomical vectors as represented in neuroimages
Functionally characterize the cerebral lobes from surface views
Functionally characterize the major structures of the diencephalon
Functionally characterize the brainstem and cerebellum
Functionally characterize the spinal cord

Organisation of the Nervous System

Functionally characterize cerebral gray matter (nuclei) and white matter (tracts)
Describe communications between the central and peripheral nervous systems
Functionally characterize the cranial nerves from surface views
Identify five major segmented parts of the spinal cord
Describe the transitionary zone between central and peripheral nervous systems
Describe the basic principles of somatic motor and sensory pathways
Explain how unilateral CNS or PNS lesions cause ipsilateral or contralateral functional deficits

Practical Clinical Neuroimaging

Describe the physics, diagnostic use and contraindications / limitations of X-ray-based techniques (excluding CT)
Describe the physics, diagnostic use and contraindications / limitations of CT
Describe the physics, diagnostic use and contraindications / limitations of MRI

Meninges & Arteries

Describe the structural relations among the cranial meninges
Distinguish the four types of intracranial hemorrhages based on blood source and appearance on neuroimage
Identify the major intracranial subarachnoid vessels and their general cerebral distributions
Identify the major subarachnoid vessels and their general bulbar distributions.
Describe the structural relations among the spinal meninges
Describe the sources and distribution of spinal blood, as depicted in horizontal sections

Subdural Hematoma

Critically appraise the symptoms and underlying pathology(ies) of subdural hematoma, the differential diagnoses and possible site(s) of the lesion and nature of the disease.

SOM.CS.III.BPM2.4.NB.4.NSCI.0477
SOM.CS.II.BPM2.4.NB.4.NSCI.0478
SOM.CS.II.BPM2.4.NB.4.NSCI.0479
SOM.CS.II.BPM2.4.NB.4.NSCI.0480
SOM.CS.II.BPM2.4.NB.4.NSCI.0481

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0010
SOM.MK.I.BPM2.1.NB.1.NSCI.0011
SOM.MK.I.BPM2.1.NB.1.NSCI.0012
SOM.MK.I.BPM2.1.NB.1.NSCI.0013
SOM.MK.I.BPM2.1.NB.1.NSCI.0014
SOM.MK.I.BPM2.1.NB.1.NSCI.0015
SOM.MK.I.BPM2.1.NB.1.NSCI.0016
SOM.MK.I.BPM2.1.NB.1.NSCI.0017
SOM.MK.I.BPM2.1.NB.1.NSCI.0018

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0061
SOM.MK.I.BPM2.1.NB.1.NSCI.0529
SOM.MK.I.BPM2.1.NB.1.NSCI.0530
SOM.MK.I.BPM2.1.NB.1.NSCI.0062
SOM.MK.I.BPM2.1.NB.1.NSCI.0531
SOM.MK.I.BPM2.1.NB.1.NSCI.0063
SOM.MK.I.BPM2.1.NB.1.NSCI.0064
SOM.MK.I.BPM2.1.NB.1.NSCI.0065
SOM.MK.I.BPM2.1.NB.2.NSCI.0066
SOM.MK.II.BPM2.1.NB.2.NSCI.0067
SOM.MK.I.BPM2.1.NB.1.NSCI.0068
SOM.MK.II.BPM2.1.NB.2.NSCI.0069

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0026
SOM.MK.I.BPM2.1.NB.1.NSCI.0027
SOM.MK.I.BPM2.1.NB.1.NSCI.0028
SOM.MK.I.BPM2.1.NB.1.NSCI.0029
SOM.MK.I.BPM2.1.NB.1.NSCI.0030
SOM.MK.I.BPM2.1.NB.1.NSCI.0031
SOM.MK.I.BPM2.1.NB.1.NSCI.0032
SOM.MK.I.BPM2.1.NB.1.NSCI.0033
SOM.MK.I.BPM2.1.NB.1.NSCI.0034
SOM.MK.I.BPM2.1.NB.1.NSCI.0035

DLA NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0036

In the context of subdural hematoma, describe the major components of a mental status examination.
In the context of subdural hematoma, describe the retinal projections to the different areas of the brain.
In the context of subdural hematoma, describe how the visual field is represented on the retina
In the context of subdural hematoma, correlate upper and lower motor neuronal signs with lesions affecting central and peripheral nervous sites
Apply knowledge of the corticospinal system to subdural hematoma.

Neurons and Glia

Identify two major cell types in the nervous system
Identify the dendrites, soma, axon and axon terminals of neurons
Describe functional properties of neuronal organelles
Correlate cytoskeletal units with active transport mechanisms
Correlate neuronal morphotype and intercellular reception/transmission
Correlate conduction velocity with axonal myelination and diameter
Differentiate excitatory and inhibitory neurons
Describe the types of glial cell and their functions in the CNS and PNS
List cellular reactions to axonal transection

Transport within the CNS

List the fluid compartments of the CNS
List the functions and constituents of the CSF
Identify the typical sites for lumbar puncture in the adult
Discuss principles of measurement and pathological elevation of intracranial pressure
Describe the blood brain barrier (BBB) and its transport mechanisms
Discuss weakening of the BBB tied to the circumventricular organs
Describe the role of the choroid plexus in CSF formation and of the arachnoid villi in CSF drainage
Describe transfer of solute and water across barriers in the central nervous system
Correlate composition of the CSF with disease processes
Identify the causes of non-communicating and communicating hydrocephalus
Describe pathways for movement of fluid and solutes among different compartments in the CNS
Identify the causes of brain edema, other than hydrocephalus

Stimulated Excitable Cells

Describe the relation between the stimulus delivered to an excitable cell and the associated graded electrical response
Explain how excitable cells convert stimuli into responses consisting of changes of conductance and membrane potential
Explain the concept of electrical signals spreading from their sites of origin with decrement
Relate the electrical properties of cable-like structures, such as axons and dendrites, to their length constants
Describe the all-or-none response to depolarization, the threshold for excitation and the physical features determining the trigger (spike-initiating) zone in neurons
Explain how the structural features of voltage-gated sodium channels are related to the influence of depolarization on the closed and open states of this channel
Localize voltage-gated channels in unmyelinated and myelinated axons
Describe how impulses are propagated in unmyelinated and myelinated axons
Relate conduction velocity to myelination and axonal diameter
Explain action-potential blocking effects of lidocaine

Excitable Cells

Describe the influence of the influx of sodium ions on the rising phase of the action potential

SOM.MK.I.BPM2.1.NB.1.NSCI.0037 Describe the influence of the efflux of potassium ions on the repolarizing phase of the action potential
SOM.MK.I.BPM2.1.NB.1.NSCI.0038 Describe the sequential changes in sodium and potassium conductance
SOM.MK.IV.BPM2.1.NB.1.NSCI.0039 Explain absolute and relative refractory periods

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0092 Describe the morphological and functional characteristics of the various types of synapses
SOM.MK.I.BPM2.1.NB.1.NSCI.0093 Contrast ionotropic and metabotropic receptors
SOM.MK.I.BPM2.1.NB.1.NSCI.0094 List five steps in the life cycles of small molecule neurotransmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0095 List five steps in the life cycle of neuroactive peptides
SOM.MK.I.BPM2.1.NB.1.NSCI.0096 List five events that underlie the interval between a presynaptic impulse and postsynaptic responses to neurotransmitter
SOM.MK.I.BPM2.1.NB.1.NSCI.0097 Describe quantal release of neurotransmitters
SOM.MK.IV.BPM2.1.NB.1.NSCI.0098 Explain the effects of select substances on release of transmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0099 Explain decrement and summation of postsynaptic potentials
SOM.MK.I.BPM2.1.NB.1.NSCI.0100 Explain mechanisms for destruction/removal of neurotransmitter from the synapse

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0111 Describe the nuclei and projections and receptors tied to acetylcholine
SOM.MK.I.BPM2.1.NB.1.NSCI.0112 Describe the nuclei and projections and receptors tied to amino acid transmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0113 Describe the nuclei and projections and receptors tied to catecholamines and serotonin

DLA NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0101 List the five traditional criteria that define a neurotransmitter
SOM.MK.I.BPM2.1.NB.1.NSCI.0102 Classify neurotransmitters according to molecular weight
SOM.MK.I.BPM2.1.NB.1.NSCI.0103 Describe mechanisms for synthesis of neurotransmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0104 Describe mechanisms for loading of transmitters into vesicles
SOM.MK.I.BPM2.1.NB.1.NSCI.0105 Describe the synthesis/removal of acetylcholine
SOM.MK.I.BPM2.1.NB.1.NSCI.0106 Describe the synthesis/removal of amino acid transmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0107 Describe the synthesis/removal of catecholamines and serotonin
SOM.MK.I.BPM2.1.NB. 1.NSCI.0109 Describe the synthesis/removal of neuroactive peptides
SOM.MK.I.BPM2.1.NB. 1.NSCI.0110 Describe the synthesis/removal of nitric oxide

Lecture ANAT
SOM.MKII.BPM2.1.NB.1.ANAT.0001 Identify the different bones and sutures of the skull
SOM.MKII.BPM2.1.NB.1.ANAT.0002 Identify the bones that constitute the viscerocranium of the skull and the foramina found in this part of the skull
SOM.MKII.BPM2.1.NB.1.ANAT.0003 Identify the bones that constitute the neurocranium and the foramina found in this part of the skull
SOM.MKII.BPM2.1.NB.1.ANAT.0004 Identify the boundaries of the anterior, middle and posterior cranial fossae
SOM.MKII.BPM2.1.NB.1.ANAT.0005 Identify the foramina of the skull from the internal and external surfaces of the skull
SOM.MKII.BPM2.1.NB.1.ANAT.0006 Describe the structures passing through each of the foramina
SOM.MKII.BPM2.1.NB.1.ANAT.0012 List the names and give the roman numerical assigned to each of the cranial nerves.
SOM.MKII.BPM2.1.NB.1.ANAT.0013 List the cranial nerves found in each of the cranial fossa.
SOM.MKII.BPM2.1.NB.1.ANAT.0014 Describe the types of nerve fibers found in each of the cranial nerves.
SOM.MKII.BPM2.1.NB.1.ANAT.0015 Give the location of the nucleus of each of the cranial nerves.
SOM.MKII.BPM2.1.NB.1.ANAT.0016 Describe a typical cranial pathway for each of the different types of innervation
SOM.MKII.BPM2.1.NB.1.ANAT.0224 Identify the different bones of the skull on X-Ray, CT and MRI imaging modalities
SOM.MKII.BPM2.1.NB.1.ANAT.0233 Identify the vessels that supply the head, neck and brain on an angiogram

Lecture ANAT

- SOM.MKII.BPM2.1.NB.1.ANAT.0064
- SOM.MKII.BPM2.1.NB.1.ANAT.0065
- SOM.MKII.BPM2.1.NB.1.ANAT.0066
- SOM.MKII.BPM2.1.NB.1.ANAT.0067
- SOM.MKII.BPM2.1.NB.1.ANAT.0068
- SOM.MKII.BPM2.1.NB.1.ANAT.0069
- SOM.MKII.BPM2.1.NB.1.ANAT.0070
- SOM.MKII.BPM2.1.NB.1.ANAT.0071
- SOM.MKII.BPM2.1.NB.1.ANAT.0072
- SOM.MKII.BPM2.1.NB.1.ANAT.0073
- SOM.MKII.BPM2.1.NB.1.ANAT.0074
- SOM.MKII.BPM2.1.NB.1.ANAT.0075
- SOM.MKII.BPM2.1.NB.1.ANAT.0076
- SOM.MKII.BPM2.1.NB.1.ANAT.0077
- SOM.MKII.BPM2.1.NB.1.ANAT.0078
- SOM.MKII.BPM2.1.NB.1.ANAT.0079
- SOM.MKII.BPM2.1.NB.1.ANAT.0080
- SOM.MKII.BPM2.1.NB.1.ANAT.0081
- SOM.MKII.BPM2.1.NB.1.ANAT.0226
- SOM.MKII.BPM2.1.NB.1.ANAT.0191
- SOM.MKII.BPM2.1.NB.1.ANAT.0192
- SOM.MKII.BPM2.1.NB.1.ANAT.0193
- SOM.MKII.BPM2.1.NB.1.ANAT.0194
- SOM.MKII.BPM2.1.NB.1.ANAT.0195

Lecture ANAT

- SOM.MKII.BPM2.1.NB.1.ANAT.0082
- SOM.MKII.BPM2.1.NB.1.ANAT.0083
- SOM.MKII.BPM2.1.NB.1.ANAT.0084
- SOM.MKII.BPM2.1.NB.1.ANAT.0085
- SOM.MKII.BPM2.1.NB.1.ANAT.0086
- SOM.MKII.BPM2.1.NB.1.ANAT.0087
- SOM.MKII.BPM2.1.NB.1.ANAT.0088
- SOM.MKII.BPM2.1.NB.1.ANAT.0089
- SOM.MKII.BPM2.1.NB.1.ANAT.0090
- SOM.MKII.BPM2.1.NB.1.ANAT.0091
- SOM.MKII.BPM2.1.NB.1.ANAT.0092
- SOM.MKII.BPM2.1.NB.1.ANAT.0093
- SOM.MKII.BPM2.1.NB.1.ANAT.0094
- SOM.MKII.BPM2.1.NB.1.ANAT.0095
- SOM.MKII.BPM2.1.NB.1.ANAT.0227
- SOM.MKII.BPM2.1.NB.1.ANAT.0225

Triangles of the Neck

- Describe the fascial layers of the neck including investing fascia, pretracheal and buccopharyngeal fascia, prevertebral fascia and alar fascia and their clinical relevance
- Describe the general attachments, innervation and actions of the muscles that define the borders of the posterior triangle
- Describe the general attachments, innervations and actions of the muscles that are found in the floor of the posterior triangle
- Describe the origin, course and distribution of the spinal accessory nerve
- Describe the borders of the occipital and subclavian triangles.
- Describe the transverse cervical artery, its branches and distribution
- Describe the cervical plexus, its branches and their distribution
- Describe the general attachments, innervation and actions of the muscles that define the borders of the anterior triangle
- Describe the general attachments, innervation and actions of the muscles that define the borders of the submental, carotid, muscular and submandibular triangles
- List the major structures found in each of the sub-triangles of the anterior triangle and explain the importance of these subdivisions
- Describe the branches of the external carotid artery and their areas of distribution
- Describe the internal carotid artery in the neck and discuss the clinical importance of the carotid sinus
- List the structures in the carotid sheath and their relationship to each other
- Describe the relationship of the carotid sheath to the prevertebral and pretracheal fasciae, the cervical part of the sympathetic trunk and the phrenic nerve
- Describe the course of the hypoglossal nerve and its relationship to the carotid arteries, C1 nerve, the hyoglossus muscle and the posterior belly of digastric
- Describe the boundaries of the scalene gap and list its contents
- Describe the location, relationships, structure, blood supply, venous and lymphatic drainage of the thyroid gland.
- Describe the venous and lymphatic drainage of the superficial and deep structures of the neck
- Identify major anatomical structures of the neck in MRI, CT and X-ray imaging
- Give the exit foramen for the accessory nerve and state which bone it is found in
- Describe the types of nerve fibers found in the accessory nerve
- Describe the branches and distribution of the accessory nerve
- Describe the extracranial course of the accessory nerve and what it supplies
- Give the location of the nuclei of the accessory nerve

Face and Scalp

- List the locations, innervations and actions of the major muscle groups of facial expression
- Describe the course of the facial artery and vein
- Describe the location, innervation and blood supply of the parotid gland
- Describe the exit of the facial nerve from the temporal bone via the stylomastoid foramen and its relationship to structures as it courses towards and through the parotid gland
- Describe the extracranial course of the facial nerve and its distribution
- Describe the clinical significance of the buccal fat pad
- Describe the course of the parotid duct and its opening in the oral vestibule
- Discuss how pathology of the parotid gland may lead to facial palsy
- Describe the five layers of the scalp, its innervation, blood supply and venous return
- Describe the “danger zone” of the scalp and its clinical significance
- Describe how infections of the scalp and the face may spread into the cranial cavity
- Describe the lymphatic drainage of the superficial and deep structures of the face and scalp
- Describe the causes and locations of caput succedaneum, cephalohematoma and subgaleal hematoma
- Describe the sensory innervation of the face
- Identify major anatomical structures of the face and scalp in MRI, CT and X-Ray imaging
- Identify the spaces of the skull on X-ray, CT and MRI modalities

Lecture ANAT	
SOM.MKII.BPM2.1.NB.1.ANAT.0157	Cranial Nerves V and VII
SOM.MKII.BPM2.1.NB.1.ANAT.0158	Describe how and where the trigeminal nerve (CNV) attaches to brainstem
SOM.MKII.BPM2.1.NB.1.ANAT.0159	Describe the locations of the three nuclei of the trigeminal nerve and their associated structures
SOM.MKII.BPM2.1.NB.1.ANAT.0160	Describe the three sensory divisions of the trigeminal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0161	Describe the motor component of the trigeminal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0162	Describe the intracranial portion of the trigeminal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0162	Describe the trigeminal ganglion, including its composition and location
SOM.MKII.BPM2.1.NB.1.ANAT.0163	Describe the foramina through which each of the divisions of the trigeminal nerve pass through
SOM.MKII.BPM2.1.NB.1.ANAT.0164	Describe the extracranial pathway of each of the divisions of the trigeminal nerve and the structures that they supply
SOM.MKII.BPM2.1.NB.1.ANAT.0165	Describe the consequence of damage to the trigeminal nerve at different locations: brainstem, in the cavernous sinus and at each foramen of exit.
SOM.MKII.BPM2.1.NB.1.ANAT.0166	Describe how and where the facial nerve (CNVII) attaches to brainstem
SOM.MKII.BPM2.1.NB.1.ANAT.0167	Describe the locations of the nuclei of the facial nerve and their associated structures
SOM.MKII.BPM2.1.NB.1.ANAT.0168	Describe the different components of the facial nerve and their origin
SOM.MKII.BPM2.1.NB.1.ANAT.0169	Describe the parasympathetic component of the facial nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0170	Describe the intracranial portion of the facial nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0171	Describe the geniculate ganglion, including its composition, location and branches
SOM.MKII.BPM2.1.NB.1.ANAT.0172	Describe the pathway the facial nerve takes to pass from the intra- to extracranial portions
SOM.MKII.BPM2.1.NB.1.ANAT.0173	Describe the extracranial pathway of the facial nerve, its branches and the structures that they supply
SOM.MKII.BPM2.1.NB.1.ANAT.0174	Describe the consequence of damage to the facial nerve at different locations: brainstem, in the facial canal and at each foramen.
Small group NSCI	NB SG 01 - CNS Structures
SOM.MK.I.BPM2.5.NB.5.NSCI.0555	Examine the five parts of the brain and the cerebellum, including their key structures (as covered in the associated lectures) on the anatomical model of a midsagittal section of the head, while relating them to the ventricular system
SOM.MK.I.BPM2.5.NB.5.NSCI.0556	Examine the five lobes of the cerebral cortex, the boundaries separating them, and their key structures (as covered in the associated lectures) on the anatomical model of half of the brain, while relating them to the ventricular system
SOM.MK.I.BPM2.5.NB.5.NSCI.0557	Examine the diencephalon and the three parts of the brainstem, the boundaries separating them, and some of their key structures (as covered in the associated lectures) on the anatomical model of the brainstem, while relating them to the ventricular system
SOM.MK.I.BPM2.5.NB.5.NSCI.0558	Examine the elements of the ventricular system (as covered in the associated lectures) on the anatomical model of the ventricular system, while relating them to the relevant structures of the central nervous system models
SOM.MK.I.BPM2.5.NB.5.NSCI.0559	Relate the functional deficits experienced by a patient in a clinical scenario to the functional anatomy of the central nervous system
Lecture ANAT	
SOM.MKII.BPM2.1.NB.1.ANAT.0142	Infratemporal and Pterygopalatine Fossae
SOM.MKII.BPM2.1.NB.1.ANAT.0143	List the bony boundaries of the infratemporal fossa
SOM.MKII.BPM2.1.NB.1.ANAT.0144	Describe the structure, innervation and movements of the temporomandibular joint
SOM.MKII.BPM2.1.NB.1.ANAT.0144	Describe the general attachments, innervations and actions of the muscles of mastication
SOM.MKII.BPM2.1.NB.1.ANAT.0145	Describe the course of the maxillary artery in the infratemporal fossa and the areas supplied by its branches
SOM.MKII.BPM2.1.NB.1.ANAT.0146	Describe the pterygoid venous plexus and list its intra-and extracranial connections
SOM.MKII.BPM2.1.NB.1.ANAT.0147	Describe the mandibular nerve, giving its entrance foramen into the fossa, its branches, the nerve fiber content and distribution of each branch
SOM.MKII.BPM2.1.NB.1.ANAT.0148	Describe the chorda tympani nerve; its origin, fiber types, connection to the lingual nerve, target ganglion and final distribution
SOM.MKII.BPM2.1.NB.1.ANAT.0149	Describe the typical site of injection of anesthetic for dental work of the mandibular and maxillary teeth
SOM.MKII.BPM2.1.NB.1.ANAT.0150	Describe the boundaries of the pterygopalatine fossa
SOM.MKII.BPM2.1.NB.1.ANAT.0151	Describe the foramina and connections of the pterygopalatine fossa
SOM.MKII.BPM2.1.NB.1.ANAT.0152	Describe the contents of the pterygopalatine fossa and the relationship of these structures to each other
SOM.MKII.BPM2.1.NB.1.ANAT.0153	Describe the pterygopalatine ganglion, the nerve fibers contributing to its formation, its branches and their distribution
SOM.MKII.BPM2.1.NB.1.ANAT.0154	List the fiber types in the nerve of the pterygoid canal and describe which nerves join to form it

SOM.MKII.BPM2.1.NB.1.ANAT.0155

SOM.MKII.BPM2.1.NB.1.ANAT.0156

SOM.MKII.BPM2.1.NB.1.ANAT.0228

Describe the pterygopalatine portion of the maxillary artery

Describe the course and distribution of the sphenopalatine and infraorbital arteries

Identify major anatomical structures of the infratemporal and pterygopalatine fossa in MRI, CT and X-ray imaging

Lecture ANAT

SOM.MKII.BPM2.1.NB.1.ANAT.0096

SOM.MKII.BPM2.1.NB.1.ANAT.0097

SOM.MKII.BPM2.1.NB.1.ANAT.0098

SOM.MKII.BPM2.1.NB.1.ANAT.0099

SOM.MKII.BPM2.1.NB.1.ANAT.0100

SOM.MKII.BPM2.1.NB.1.ANAT.0101

SOM.MKII.BPM2.1.NB.1.ANAT.0102

SOM.MKII.BPM2.1.NB.1.ANAT.0107

SOM.MKII.BPM2.1.NB.1.ANAT.0229

Nasal Cavity

List the bones that form the boundaries of the nasal cavity

List the bones and cartilage that form the nasal septum

Describe the sensory and parasympathetic innervation of the nasal cavity

Describe the structure and function of the turbinates

Describe the blood supply and lymphatic drainage of the nasal cavity

Describe the paranasal sinuses, the bones in which they are found, and where each opens into the nasal cavity

Describe the sensory and parasympathetic innervations of the paranasal sinuses.

Explain the clinical significance of Kiesselbach’s area in relation to epistaxis

Identify major anatomical structures of the oral and nasal regions in MRI, CT and X-ray imaging

Lecture ANAT;NSCI

SOM.MKII.BPM2.1.NB.1.ANAT.0124

SOM.MKII.BPM2.1.NB.1.ANAT.0125

SOM.MKII.BPM2.1.NB.1.ANAT.0126

SOM.MKII.BPM2.1.NB.1.ANAT.0064

SOM.MKII.BPM2.1.NB.1.ANAT.0128

SOM.MKII.BPM2.1.NB.1.ANAT.0129

SOM.MKII.BPM2.1.NB.1.ANAT.0130

SOM.MKII.BPM2.1.NB.1.ANAT.0131

SOM.MKII.BPM2.1.NB.1.ANAT.0255

SOM.MKII.BPM2.1.NB.1.ANAT.0132

SOM.MKII.BPM2.1.NB.1.ANAT.0133

SOM.MKII.BPM2.1.NB.1.ANAT.0134

SOM.MKII.BPM2.1.NB.1.ANAT.0135

SOM.MKII.BPM2.1.NB.1.ANAT.0136

SOM.MKII.BPM2.1.NB.1.ANAT.0137

SOM.MKII.BPM2.1.NB.1.ANAT.0138

SOM.MKII.BPM2.1.NB.1.ANAT.0139

SOM.MKII.BPM2.1.NB.1.ANAT.0141

SOM.MKII.BPM2.1.NB.1.ANAT.1168

SOM.MKII.BPM2.1.NB.1.ANAT.0230

Pharynx and Larynx

Delineate the borders between the naso-, oro-, and laryngo-pharynx

List the general attachments, innervations and functions of the muscles of the pharynx

List the contributions to the pharyngeal nerve plexus

Describe the fascial layers of the neck including investing fascia, pretracheal and buccopharyngeal fascia, prevertebral fascia and alar fascia and their clinical relevance

Explain the sensory and motor innervation of the nasopharynx, oropharynx, and laryngopharynx

Describe the piriform fossa, its innervations, and clinical importance

Describe the arterial supply of the pharynx, and its venous and lymphatic drainage

Describe the location of the pharyngeal, palatine, and lingual tonsils

Describe the clinical importance of the arrangement of pharyngeal, palatine and lingual tonsils

Identify the hyoid bone and the cartilages that form the larynx

Identify the thyrohyoid and cricothyroid membranes

Define the boundaries and extent of the larynx

Describe the true and false vocal cords, ventricle, vestibule, and rima glottidis

List the attachments, innervation and functions of the muscles of the larynx

Describe the sensory and mucus innervation of the larynx

Describe the arterial supply of the larynx, and its venous and lymphatic drainage

Compare and contrast the clinical picture of a patient with a loss of recurrent versus internal versus external laryngeal nerve

Explain how the pharynx and larynx function during deglutition

Compare and contrast structures incised during a tracheotomy versus a cricothyrotomy procedure

Identify major anatomical structures of the pharynx in MRI, CT and X-ray imaging

Lecture ANAT:NSCI

SOM.MKII.BPM2.1.NB.1.ANAT.0175

SOM.MKII.BPM2.1.NB.1.ANAT.0176

SOM.MKII.BPM2.1.NB.1.ANAT.0177

SOM.MKII.BPM2.1.NB.1.ANAT.0178

SOM.MKII.BPM2.1.NB.1.ANAT.0179

SOM.MKII.BPM2.1.NB.1.ANAT.0180

SOM.MKII.BPM2.1.NB.1.ANAT.0181

Cranial Nerves IX and X

List the names and give the roman numerical assigned to cranial nerves nine to twelve

Describe the boundaries of the posterior cranial fossa and the positioning of the foramina

Give the exit foramen for the glossopharyngeal nerve and state which bone it is found in

Describe the types of nerve fibers found in the glossopharyngeal nerve

Describe the branches and distribution of the glossopharyngeal nerve

Describe the extracranial course of the glossopharyngeal nerve and what it supplies

Give the location of the nuclei of the glossopharyngeal nerve

SOM.MKII.BPM2.1.NB.1.ANAT.0182	Give the name and location of the ganglia formed by the glossopharyngeal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0183	Explain the consequence of extracranial lesions of the glossopharyngeal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0184	Give the exit foramen for the vagus nerve and state which bone it is found in
SOM.MKII.BPM2.1.NB.1.ANAT.0185	Describe the types of nerve fibers found in the vagus nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0186	Describe the branches and distribution of the vagus nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0187	Describe the extracranial course of the vagus nerve and what it supplies
SOM.MKII.BPM2.1.NB.1.ANAT.0188	Give the location of the nuclei of the vagus nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0189	Give the name and location of the ganglia formed by the vagus nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0190	Explain the consequence of extracranial lesions of the vagus nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0203	Describe the afferent and efferent limbs of the following reflexes: cough and gag
Small group NSCI	NB SG 02 - Functional Morphology Examine the diencephalon, the three parts of the brainstem, and the cerebellum, the boundaries separating them, and their key surface structures (as covered in the associated lectures) on the anatomical brainstem and sections of the head models, while relating them to the ventricular system Relate the functional deficits experienced by a patient in a clinical scenario to the functional anatomy of brainstem and cerebellum Correlate the knowledge of brainstem origin and function of cranial nerves with localizing lesions based on signs and symptoms Correlate the knowledge of spinal cord segmentation and spinal nerve origin and function with localizing lesions based on signs and symptoms
Lecture ANAT	Oral Cavity and Palate Describe the structure and function of the hard and soft palate Describe the bones that form the hard palate Describe the mucosal coverings and innervation of the hard palate Describe the muscles, function and innervations of the soft palate Describe the boundaries of the oral cavity and its division into the oral cavity proper and oral vestibule Describe the sensory innervation of the oral vestibule Describe the sensory innervation of the oral cavity proper Describe the course of the submandibular duct and its termination under the tongue Describe the location of the sublingual gland and the openings of this gland Differentiate between the oral and pharyngeal parts of the tongue List the nerves that carry general sensation and taste sensation from the tongue List the general attachments, innervations and actions of the extrinsic muscles of the tongue Describe how the tongue and palate function during deglutition List the general attachments, innervations and actions of the muscles that are found in the floor of the mouth Describe the arterial supply and venous and lymphatic drainage of the tongue Give the exit foramen for the hypoglossal nerve and state which bone it is found in Describe the types of nerve fibers found in the hypoglossal nerve Describe the branches and distribution of the hypoglossal nerve Describe the extracranial course of the hypoglossal nerve and what it supplies Give the location of the nuclei of the hypoglossal nerve Explain the consequence of extracranial lesions of the hypoglossal nerve
DLA ANAT	Chemical Senses - Gustation Describe the basic taste qualities, and explain the functional anatomy of the gustatory system, focusing on taste buds and receptor cells, and the different signal transduction processes Describe the gustatory pathways including the location of cortical gustatory areas Contrast the functional anatomy of the ageusia, anosmia, olfactory hallucinations in epileptic seizures.

DLA ANAT
SOM.CS.I.BPM2.1.NB.4.NSCI.0268
SOM.MK.I.BPM2.1.NB.4.NSCI.0271

Lecture ANAT
SOM.MKII.BPM2.1.NB.1.ANAT.0017
SOM.MKII.BPM2.1.NB.1.ANAT.0018
SOM.MKII.BPM2.1.NB.1.ANAT.0019
SOM.MKII.BPM2.1.NB.1.ANAT.0020
SOM.MKII.BPM2.1.NB.1.ANAT.0021
SOM.MKII.BPM2.1.NB.1.ANAT.0023
SOM.MKII.BPM2.1.NB.1.ANAT.0022
SOM.MKII.BPM2.1.NB.1.ANAT.0024
SOM.MKII.BPM2.1.NB.1.ANAT.0025
SOM.MKII.BPM2.1.NB.1.ANAT.0026
SOM.MKII.BPM2.1.NB.1.ANAT.0027
SOM.MKII.BPM2.1.NB.1.ANAT.0028
SOM.MKII.BPM2.1.NB.1.ANAT.0029
SOM.MKII.BPM2.1.NB.1.ANAT.0030
SOM.MKII.BPM2.1.NB.1.ANAT.0033
SOM.MKII.BPM2.1.NB.1.ANAT.0031
SOM.MKII.BPM2.1.NB.1.ANAT.0032
SOM.MKII.BPM2.1.NB.1.ANAT.0247
SOM.MKII.BPM2.1.NB.1.ANAT.0231

Lecture HCB
SOM.MK.I.BPM2.4.NB.1.HCB.1329
SOM.MK.I.BPM2.4.NB.1.HCB.1330
SOM.MK.I.BPM2.4.NB.1.HCB.1331
SOM.MK.I.BPM2.4.NB.1.HCB.1332
SOM.MK.I.BPM2.4.NB.1.HCB.1333
SOM.MK.I.BPM2.4.NB.1.HCB.1334
SOM.MK.I.BPM2.4.NB.1.HCB.1335
SOM.MK.I.BPM2.4.NB.1.HCB.1336
SOM.MK.I.BPM2.4.NB.1.HCB.1337
SOM.MK.I.BPM2.4.NB.1.HCB.1338
SOM.MK.I.BPM2.4.NB.1.HCB.1339
SOM.MK.I.BPM2.4.NB.1.HCB.1340
SOM.MK.I.BPM2.4.NB.1.HCB.1341
SOM.MK.I.BPM2.4.NB.1.HCB.1342
SOM.MK.I.BPM2.4.NB.1.HCB.1343
SOM.MK.I.BPM2.4.NB.1.HCB.1344
SOM.MK.I.BPM2.4.NB.1.HCB.1345
SOM.MK.I.BPM2.4.NB.1.HCB.1346

Chemical Senses - Neurologic Examination

Describe neurologic examination of olfaction and gustation.
Contrast the functional anatomy of the ageusia, anosmia, olfactory hallucinations in epileptic seizures.

Gross Anatomy and Development of the Orbit and Eye

List the bones forming the orbit and the orbital margin
Describe the lacrimal apparatus and how tears flow from the lacrimal gland to the medial canthus of the eye into the nose
List the structures that pass through the optic canal and the superior orbital fissure to enter or leave the orbit
Describe the attachments, innervations and functions of the six muscles of the eye
Compare and contrast the attachments, innervations and functions of the levator palpabrae superioris and superior tarsal muscles
Describe the branches of the ophthalmic nerve in the orbit and their distribution
Describe the anatomy of the eyes including the layers of the eyeball
Describe the nerves that carry general sensation from the eyeball
Describe the course of parasympathetic fibers from the brainstem to the eye
Describe the pathway of sympathetic fibers from their origin to the dilator pupillae muscle and the superior tarsal muscle
Describe the intraorbital course of the ophthalmic artery and name its branches and distribution
Describe how the ophthalmic veins communicate with the pterygoid venous plexus, facial vein and the cavernous sinus
Describe how the extraocular muscles function in the H test
Describe the different causes of diplopia (SG)
Describe the development of congenital retinal detachment and congenital cataracts
Describe the development of the structures of the eye
Describe the development of coloboma of the iris and retina
List the nerves and muscles that may be affected by the loss of accommodation
Identify major anatomical structures of the orbit and the acoustic meatus in MRI, CT and X-Ray imaging

Histology of the Eye

List the 3 layers of the walls of the eyeball
Differentiate between the anterior and posterior chambers of the eye based on location.
Give the function(s) of the sclera of the corneoscleral coat.
Describe the microscopic structures of the sclera of the corneoscleral coat.
Give the function(s) of the cornea of the corneoscleral coat.
Describe the microscopic structures of the cornea of the corneoscleral coat.
Give the function(s) of the choroid of the uveal tract.
Describe the microscopic structure of the choroid of the uveal tract.
Describe the functional significance of choriocapillary layer of choroid
Give the function(s) of the ciliary body of the uveal tract.
Describe the microscopic structure of the ciliary body of the uveal tract.
Give the function(s) of the iris of the uveal tract.
Describe the microscopic structure of the iris of the uveal tract.
Describe the synthesis of aqueous humor.
Describe the pathway for the drainage of aqueous humor
Describe the blood supply of the retina.
Describe the microscopic structure of the retina.
Describe the microscopic structure of optic disc.

SOM.MK.I.BPM2.4.NB.1.HCB.1347
SOM.MK.I.BPM2.4.NB.1.HCB.1348
SOM.MK.I.BPM2.4.NB.1.HCB.1349
SOM.MK.I.BPM2.4.NB.1.HCB.1350
SOM.MK.I.BPM2.4.NB.1.HCB.1351
SOM.MK.I.BPM2.4.NB.1.HCB.1352
SOM.MK.I.BPM2.4.NB.1.HCB.1353
SOM.MK.I.BPM2.4.NB.1.HCB.1354
SOM.MK.I.BPM2.4.NB.1.HCB.1355
SOM.MK.I.BPM2.4.NB.1.HCB.1356
SOM.MK.I.BPM2.4.NB.1.HCB.1357
SOM.MK.I.BPM2.4.NB.1.HCB.1358

Lecture ANAT

SOM.MKII.BPM2.1.NB.1.ANAT.0035
SOM.MKII.BPM2.1.NB.1.ANAT.0036
SOM.MKII.BPM2.1.NB.1.ANAT.0037
SOM.MKII.BPM2.1.NB.1.ANAT.0038
SOM.MKII.BPM2.1.NB.1.ANAT.0039
SOM.MKII.BPM2.1.NB.1.ANAT.0040
SOM.MKII.BPM2.1.NB.1.ANAT.0041
SOM.MKII.BPM2.1.NB.1.ANAT.0042
SOM.MKII.BPM2.1.NB.1.ANAT.0043
SOM.MKII.BPM2.1.NB.1.ANAT.0044
SOM.MKII.BPM2.1.NB.1.ANAT.0045
SOM.MKII.BPM2.1.NB.1.ANAT.0046
SOM.MKII.BPM2.1.NB.1.ANAT.0047
SOM.MKII.BPM2.1.NB.1.ANAT.0048
SOM.MKII.BPM2.1.NB.1.ANAT.0049
SOM.MKII.BPM2.1.NB.1.ANAT.0050
SOM.MKII.BPM2.1.NB.1.ANAT.0051

Small group NSCI

SOM.MK.I.BPM2.5.NB.5.NSCI.0354
SOM.MK.I.BPM2.5.NB.5.NSCI.0355
SOM.MK.I.BPM2.5.NB.5.NSCI.0356
SOM.MK.I.BPM2.5.NB.5.NSCI.0357
SOM.MK.I.BPM2.5.NB.5.NSCI.0358
SOM.MK.I.BPM2.5.NB.5.NSCI.0359
SOM.MK.I.BPM2.5.NB.5.NSCI.0360
SOM.MK.I.BPM2.5.NB.5.NSCI.0361

Lecture HCB

SOM.MK.I.BPM2.4.NB.1.HCB.1359
SOM.MK.I.BPM2.4.NB.1.HCB.1360

Describe the microscopic structure of the fovea centralis.
State the function of the lens.
Describe the microscopic structure of the lens.
Give the function of the vitreous body.
Describe the microscopic structure of vitreous body.
Differentiate between the glands of the eyelids based on their microscopic structure, location and clinical significance.
Describe the microscopic structure of the conjunctiva.
Describe the structural changes in the retina in retinal detachment.
Explain the effect of multiple sclerosis on the optic nerve.
Explain the clinical significance of corneal transplant.
Describe the changes in the lens in cataract.
Describe the changes in the eye that result in glaucoma.

Gross Anatomy and Development of the Ear

Describe the anatomy of the external ear
Describe the major innervation of the auricle and the external ear canal
Name the parts of the bone in which the three ear cavities are located
Dscribe the innervation of the external and internal surfaces of the tympanic membrane and explain the ear—cough reflex
Describe the boundaries of the middle ear
List the structures found in the middle ear and their relationship to one another
List the structures found in the medial wall of the middle ear
Describe the function of the ossicles
Describe the development of the external, internal, and middle ears
Describe the development of the auricular appendages, microtia, preauricular sinus, and atresia of the external acoustic meatus
Describe the attachments, innervation and actions of the tensor tympani and stapedius muscles
Describe the connections between the nasopharynx, pharyngotympanic tube, middle ear cavity, and the mastoid air cells
List the nerve fiber types in the tympanic nerve and the composition of the tympanic nerve plexus
List the areas that receive sensory fibers from the tympanic nerve and tympanic plexus
List the nerve fiber types in the chorda tympani nerve and its relation to the ossicles and tympanic membrane
List the components of the bony labyrinth
List the nerve fiber components of the vestibulocochlear nerve and their distribution.

NB SG 03 - Meninges, Arteries and Veins

Examine the site of production and flow of cerebrospinal fluid on anatomical models of the brain
Associate the pituitary gland, pineal gland and area postrema on the anatomical model of a mid-sagittal section of the head with their functional aspects
Examine the structural organization of the meninges
Examine the structural organization of the subarachnoid space
Appraise the external features of the intracranial dura
Differentiate the intracranial dura and spinal cord dura
Examine the subarachnoid arteries
Examine the structural organization of the system of veins and venous sinuses associated with the brain

Histology of the Ear

List the overall functions of the divisions of the ear
Discuss the divisions of the ear (external, middle and internal) including their structural components.

SOM.MK.I.BPM2.4.NB.1.HCB.1361	State the function of the tympanic membrane
SOM.MK.I.BPM2.4.NB.1.HCB.1362	Describe the histological structure of the tympanic membrane
SOM.MK.I.BPM2.4.NB.1.HCB.1363	Compare endolymph vs. perilymph in terms of their composition and location
SOM.MK.I.BPM2.4.NB.1.HCB.1364	Identify the semicircular canals, vestibule and cochlear canal of the bony labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1365	List the components of the membranous labyrinth (semicircular ducts, utricle and saccule, cochlear duct)
SOM.MK.I.BPM2.4.NB.1.HCB.1366	List the two broad divisions of the membranous labyrinth (vestibular and cochlear)
SOM.MK.I.BPM2.4.NB.1.HCB.1367	List the sensory regions associated with the vestibular part of the membranous labyrinth (crista ampullaris, maculae)
SOM.MK.I.BPM2.4.NB.1.HCB.1368	Describe the histological structure of the hair cells associated with the sensory regions of the vestibular division of the membranous labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1369	Describe the two types of hair cells of the vestibular labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1370	Describe the general function of hair cells in the ear including how they convert mechanical vibrations into electrical signals involved in sound perception
SOM.MK.I.BPM2.4.NB.1.HCB.1371	State the function of the crista ampullaris
SOM.MK.I.BPM2.4.NB.1.HCB.1372	Describe the histological structure of the crista ampullaris
SOM.MK.I.BPM2.4.NB.1.HCB.1373	Explain vertigo resulting from damage to the crista ampullares
SOM.MK.I.BPM2.4.NB.1.HCB.1374	State the function of the maculae of the saccule and utricle
SOM.MK.I.BPM2.4.NB.1.HCB.1375	Describe the histological structure of the maculae of the saccule and utricle
SOM.MK.I.BPM2.4.NB.1.HCB.1376	Explain vertigo due to damage to the maculae of the saccule and utricle
SOM.MK.I.BPM2.4.NB.1.HCB.1377	Give the function of the cochlea duct.
SOM.MK.I.BPM2.4.NB.1.HCB.1378	State the sensory region associated with the cochlea part of the membranous labyrinth (Organ of Corti)
SOM.MK.I.BPM2.4.NB.1.HCB.1379	Describe the two types of hair cells of the cochlear labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1380	Describe the histological structure of the hair cells associated with the sensory regions of the cochlear division of the membranous labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1381	Describe the microscopic structure of the cochlea duct.
SOM.MK.I.BPM2.4.NB.1.HCB.1382	State the function of the Organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1383	Identify the location of the Organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1384	Describe the histological structure of the Organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1385	Explain the hearing loss due to damage to organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1386	Explain the modality of sound conduction through the ear

Lecture ANAT

SOM.MK.I.BPM2.1.NB.1.ANAT.0204	Development of the Skull, Face and Paranasal Sinuses
SOM.MK.I.BPM2.1.NB.1.ANAT.0205	Describe the development of the viscerocranium and neurocranium
SOM.MK.I.BPM2.1.NB.1.ANAT.0206	Describe the closure of the sutures of the skull
SOM.MK.I.BPM2.1.NB.1.ANAT.0211	Describe the development of scaphocephaly, brachycephaly, plagiocephaly, trigonocephaly and cranium bifidum
SOM.MK.I.BPM2.1.NB.1.ANAT.0212	Describe the development of the face and palate from the five primordial prominences
SOM.MK.I.BPM2.1.NB.1.ANAT.0213	Describe the different forms of cleft lip and palate and the developmental processes that fail leading to these defects
SOM.MK.I.BPM2.1.NB.1.ANAT.0214	Describe the development of the nasal cavity
	Describe the development of the paranasal sinuses.

DLA ANAT

SOM.MK.II.BPM2.1.NB.1.ANAT.0007	Pharyngeal Arches
SOM.MK.II.BPM2.1.NB.1.ANAT.0008	Describe the formation of the pharyngeal arches, grooves and pouches
SOM.MK.II.BPM2.1.NB.1.ANAT.0009	Describe the components of the pharyngeal apparatus
SOM.MK.II.BPM2.1.NB.1.ANAT.0010	Describe the structures derived from each pharyngeal arch, pouch and cleft/groove
SOM.MK.II.BPM2.1.NB.1.ANAT.0011	Describe the cartilage, nerve and blood vessel of each pharyngeal arch
	Describe the development of the stomodeum

Small group ANAT

SOM.MK.II.BPM1.1.NB.1.ANAT.5005	NB SG 04 - Cases of Deep Cervicofacial Spaces
	Analyze the general attachments, innervations, and actions of the muscles of mastication, evaluating their roles in the process of chewing and jaw movement.

SOM.MK.II.BPM1.1.NB.1.ANAT.5006
SOM.MK.II.BPM1.1.NB.1.ANAT.5007
SOM.MK.II.BPM1.1.NB.1.ANAT.5008
SOM.MK.II.BPM1.1.NB.1.ANAT.5009
SOM.MK.II.BPM1.1.NB.1.ANAT.5010
SOM.MK.II.BPM1.1.NB.1.ANAT.5011
SOM.MK.II.BPM1.1.NB.1.ANAT.5012
SOM.MK.II.BPM1.1.NB.1.ANAT.5013
SOM.MK.II.BPM1.1.NB.1.ANAT.5014
SOM.MK.II.BPM1.1.NB.1.ANAT.5015
SOM.MK.II.BPM1.1.NB.1.ANAT.5016
SOM.MK.II.BPM1.1.NB.1.ANAT.5017
SOM.MK.II.BPM1.1.NB.1.ANAT.5018
SOM.MK.II.BPM1.1.NB.1.ANAT.5020
SOM.MK.II.BPM1.1.NB.1.ANAT.5021
SOM.MK.II.BPM1.1.NB.1.ANAT.5022

Small group ANAT

SOM.MK.II.BPM1.1.NB.1.ANAT.5001
SOM.MK.II.BPM1.1.NB.1.ANAT.5002
SOM.MK.II.BPM1.1.NB.1.ANAT.5003
SOM.MK.II.BPM1.1.NB.1.ANAT.5004
SOM.MK.II.BPM1.1.NB.1.ANAT.5023
SOM.MK.II.BPM1.1.NB.1.ANAT.5024
SOM.MK.II.BPM1.1.NB.1.ANAT.5025
SOM.MK.II.BPM1.1.NB.1.ANAT.5026
SOM.MK.II.BPM1.1.NB.1.ANAT.5027
SOM.MK.II.BPM1.1.NB.1.ANAT.5028
SOM.MK.II.BPM1.1.NB.1.ANAT.5029
SOM.MK.II.BPM1.1.NB.1.ANAT.5030
SOM.MK.II.BPM1.1.NB.1.ANAT.5031

Lecture ANAT

SOM.MK.I.BPM2.1.NB.1.ANAT.0207
SOM.MK.I.BPM2.1.NB.1.ANAT.0208
SOM.MK.I.BPM2.1.NB.1.ANAT.0209
SOM.MK.I.BPM2.1.NB.1.ANAT.0210
SOM.MK.I.BPM2.1.NB.1.ANAT.0215
SOM.MK.I.BPM2.1.NB.1.ANAT.0216
SOM.MK.I.BPM2.1.NB.1.ANAT.0217
SOM.MK.I.BPM2.1.NB.1.ANAT.0218
SOM.MK.I.BPM2.1.NB.1.ANAT.0219
SOM.MK.I.BPM2.1.NB.1.ANAT.0220
SOM.MK.I.BPM2.1.NB.1.ANAT.0221
SOM.MK.I.BPM2.1.NB.1.ANAT.0222

Examine the mandibular nerve, identifying its entrance foramen into the fossa, its branches, the nerve fiber content, and the distribution of each branch, while assessing their clinical relevance.
Describe how the ophthalmic veins communicate with the pterygoid venous plexus, facial vein, and the cavernous sinus, analyzing the implications for venous drainage and clinical conditions.
Explain the chorda tympani nerve, discussing its origin, fiber types, connection to the lingual nerve, target ganglion, and final distribution, and evaluating its function in taste and parasympathetic innervation.
Analyze the attachment of the facial nerve (CNVII) to the brainstem, and evaluate its relationship with surrounding neural structures.
Evaluate the functional significance of the facial nerve nuclei and their associated structures, considering their roles in motor and sensory processes.
Compare the different components of the facial nerve, including their origins, and synthesize how these contribute to its overall function.
Analyze the parasympathetic component of the facial nerve, assessing its physiological impact and integration within autonomic systems.
Assess the consequences of facial nerve damage at various anatomical sites (brainstem, facial canal, and foramina), and propose clinical outcomes based on these lesions.
Analyze the general attachments, innervation, and actions of the muscles that define the borders of the posterior triangle, evaluating their functional significance in head and neck movements.
Examine the cervical plexus, identifying its branches and discussing their distribution and clinical relevance.
Describe the location, innervation, and blood supply of the parotid gland, analyzing their roles in salivary secretion and overall gland function.
Discuss how pathology of the parotid gland may lead to facial palsy, analyzing the mechanisms behind nerve involvement and assessing potential clinical outcomes.
Analyze and compare the attachment points, innervations, and functions of the levator palpebrae superioris and superior tarsal muscles.
Compare and contrast the sensory functions carried by the tympanic nerve, chorda tympani, and the vestibulocochlear nerve, and analyze how these nerves contribute to both hearing and the ear-cough reflex.
Analyze the tympanic nerve plexus, detailing its fiber types and distributions, while integrating its role in the sensory innervation of the middle ear structures and cochlear function.
Evaluate the anatomical features, innervation, and actions of the tensor tympani and stapedius muscles.

NB SG 05 - Sinonasal and Parapharyngeal Cases

Analyze the blood supply and lymphatic drainage of the nasal cavity, evaluating their roles in maintaining nasal health and function.
Examine the paranasal sinuses, identifying the bones in which they are located, and explaining where each sinus opens into the nasal cavity.
Describe the sensory and parasympathetic innervation of the nasal cavity, evaluating their contribution to nasal sensation and function.
Explain the clinical significance of Kiesselbach’s area in relation to epistaxis, discussing its role in nasal bleeding and common causes of hemorrhage.
Differentiate the functional components and pathways of the glossopharyngeal and vagus nerves based on their extracranial courses and fiber types.
Compare and contrast the nuclei and ganglia associated with the glossopharyngeal and vagus nerves, relating their anatomical positions to their functional roles.
Analyze how the distribution and branching patterns of the glossopharyngeal and vagus nerves contribute to their respective clinical presentations when damaged.
Appraise the role of each nerve's ganglia in sensory transmission and autonomic regulation, particularly in relation to reflex arcs.
Evaluate the sensory and motor innervation of the nasopharynx, oropharynx, and laryngopharynx, assessing their contributions to functions such as swallowing, speech, and respiration.
Identify and analyze the contributions to the pharyngeal nerve plexus, exploring how each component integrates into the overall pharyngeal innervation.
Examine the general attachments, innervations, and functions of the muscles of the pharynx, synthesizing how these elements coordinate to facilitate swallowing and airway protection.
Describe the sensory and mucous innervation of the larynx, explaining how these neural pathways contribute to laryngeal function, such as voice production and protection from aspiration.
Analyze the attachments, innervations, and functions of the muscles of the larynx, evaluating their role in controlling airflow, voice production, and airway protection.

Pharyngeal Apparatus Derivatives; Neural Tube Defects

Describe the structures formed by each pharyngeal arch/groove/pouch that contributes to the formation of the face
Describe the signs and symptoms of first arch syndrome
Describe the development of the branchial sinuses, fistulas and cysts
Describe the defects that can develop due to abnormal development of the grooves and pouches
Describe the formation of the tongue, give the different arches that contribute and relate this to its innervation
Describe the lymph drainage of the tongue and how it links to embryological development
Describe the development of normal and ectopic thyroid and parathyroid glands
Describe the development of a thyroglossal duct cyst and its clinical presentation
Describe the development of the pituitary gland
Describe the development of craniopharyngioma
Describe the development of meroencephaly, anencephaly microcephaly and holoprosencephaly
Describe the development of the Arnold-Chiari malformation

SOM.MK.I.BPM2.1.NB.1.ANAT.0223	Compare and contrast the cause of communicating versus noncommunicating hydrocephalus
Small group HCB SOM.MK.I.BPM2.4.NB.1.HCB.1387 SOM.MK.I.BPM2.4.NB.1.HCB.1388 SOM.MK.I.BPM2.4.NB.1.HCB.1389 SOM.P.I.BPM2.5.NB.5.HCB.1390	NB SG 06 - NB Histology of Special Senses (Eye and Ear) Identify and describe the cells, tissues, organs and associated structures of the eye and ear. Present and demonstrate knowledge of the cells, tissues, organs and associated structures of the eye and ear to the group. Apply the histology of the eye and ear to clinical cases and correlations. Enhance the development of professional skills and presentation skills.
Lecture ANAT SOM.MK.II.BPM2.1.NB.1.ANAT.0242 SOM.MK.II.BPM2.1.NB.1.ANAT.0244 SOM.MK.II.BPM2.1.NB.1.ANAT.0245 SOM.MK.II.BPM2.1.NB.1.ANAT.0248 SOM.MK.II.BPM2.1.NB.1.ANAT.0249 SOM.MK.II.BPM1.1.NB.1.ANAT.0025 SOM.MK.II.BPM2.1.NB.1.ANAT.0250 SOM.MK.II.BPM2.1.NB.1.ANAT.0251	Flipped Class: Clinical Cases and Imaging of Cranial Nerves III, V and VII Describe the cause of cavernous sinus syndrome, the nerves that may be affected and what consequences can occur Describe the intracranial position of the cranial nerves and how they are associated to nearby blood vessels and their aneurysms. Describe the causes and locations of intracranial hematomas including epidural hematoma, subdural hematoma and subarachnoid hemorrhage List the nerves and muscles involved in ptosis and partial ptosis of the eyelid Describe the signs and symptoms of facial paralysis, Bell’s palsy, Horner’s syndrome, trigeminal neuralgia Evaluate the course of parasympathetic and sympathetic fibers from the brainstem to the eye.. Describe the signs and symptoms of (shingles) herpes zoster Describe the consequence of intracranial lesions of each of the 12 cranial nerves, their nuclei and their ganglia
Lecture ANAT SOM.MK.II.BPM2.1.NB.1.ANAT.0064 SOM.MK.II.BPM2.1.NB.1.ANAT.0237 SOM.MK.II.BPM2.1.NB.1.ANAT.0235 SOM.MK.II.BPM2.1.NB.1.ANAT.0239 SOM.MK.II.BPM2.1.NB.1.ANAT.0240 SOM.MK.II.BPM2.1.NB.1.ANAT.0241 SOM.MK.II.BPM2.1.NB.1.ANAT.0243 SOM.MK.II.BPM2.1.NB.1.ANAT.0246 SOM.MK.II.BPM2.1.NB.1.ANAT.0245 SOM.MK.II.BPM2.1.NB.1.ANAT.0190 SOM.MK.I.BMP2.1.NB.1.ANAT.0209 SOM.MK.I.BMP2.1.NB.1.ANAT.0218	Flipped Class: Clinical Cases and Imaging of the Head and Neck Describe the fascial layers of the neck including investing fascia, pretracheal and buccopharyngeal fascia, prevertebral fascia and alar fascia and their clinical relevance Describe the retropharyngeal space and its extension to the posterior mediastinum Describe the third (danger) space and its extension to the through the posterior mediastinum to the diaphragm Describe the anatomical relationships that are important for successful insertion of a catheter for central venous access and the possible complications of central venous access Describe fractures of the orbit and their clinical consequences including extraocular muscle impingement Describe the types of le forte facial fractures and the clinical consequences Describe the nerves responsible for mediation of pain sensation during headaches Describe the anatomical location, mechanism and clinical sequences of tentorial herniation Describe the causes and locations of intracranial hematomas including epidural hematoma, subdural hematoma and subarachnoid hemorrhage Explain the consequence of extracranial lesions of the vagus nerve Describe the development of the branchial sinuses, fistulas, and cysts Describe the development of a thyroglossal duct cyst and its clinical presentation
Lecture NSCI SOM.MK.I.BPM2.1.NB.1.NSCI.0563 SOM.MK.I.BPM2.1.NB.1.NSCI.0564 SOM.MK.I.BPM2.1.NB.1.NSCI.0565 SOM.MK.I.BPM2.1.NB.1.NSCI.0566 SOM.MK.I.BPM2.1.NB.1.NSCI.0567 SOM.MK.I.BPM2.1.NB.1.NSCI.0568 SOM.MK.I.BPM2.1.NB.1.NSCI.0569 SOM.MK.I.BPM2.1.NB.1.NSCI.0570	Section Anatomy of the CNS Describe the basic principles on which the major neuroimaging techniques, Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) are based (Required Reading - Siegel and Sapru Chapter 27 Vascular Syndromes, Computed Tomography and Magnetic Resonance Imaging) Describe the normal appearance of white matter, gray matter, cerebrospinal fluid, bone and fat on CT, MRI-T1 and MRI-T2 neuroimages (Required Reading) Describe the pathological appearance of ischemia, hemorrhage and enhanced tumor on CT, MRI-T1 and MRI-T2 neuroimages (Required Reading) Relate the key surface structures of the central nervous system and the cerebrospinal fluid space to their appearance on neuroimages Describe the key nuclei and fiber tracts on CNS sections in anatomical and clinical orientation, as well as the appropriate related anatomical vectors Describe the primary functions of the key structures identified on schematic illustrations of sections of the central nervous system Apply the knowledge of the surface location of subarachnoid arteries in describing the functional consequences of loss of blood supply to the internal structures Apply knowledge of central nervous system section anatomy and function to the process of clinical reasoning
Lecture NSCI SOM.MK.I.BPM2.1.NB.1.NSCI.0114	Somatosensory System Describe the morphology, location and functional characteristics of somatosensory receptors of the skin and of the skeleto-muscular system

SOM.MK.I.BPM2.1.NB.1.NSCI.0115
SOM.MK.I.BPM2.4.NB.4.NSCI.0116

SOM.MK.I.BPM2.1.NB.1.NSCI.0117
SOM.MK.I.BPM2.4.NB.4.NSCI.0118
SOM.MK.I.BPM2.4.NB.4.NSCI.0119

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0456
SOM.CS.III.BPM2.4.NB.4.NSCI.0457
SOM.CS.III.BPM2.4.NB.4.NSCI.0458

SOM.CS.II.BPM2.4.NB.4.NSCI.0459

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0128
SOM.MK.I.BPM2.4.NB.4.NSCI.0129
SOM.MK.I.BPM2.1.NB.1.NSCI.0130
SOM.MK.I.BPM2.4.NB.4.NSCI.0131
SOM.MK.I.BPM2.1.NB.1.NSCI.0132
SOM.CS.I.BPM2.4.NB.4.NSCI.0133
SOM.MK.I.BPM2.4.NB.4.NSCI.0134

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0120
SOM.MK.I.BPM2.1.NB.1.NSCI.0121
SOM.MK.I.BPM2.4.NB.4.NSCI.0122
SOM.MK.I.BPM2.1.NB.1.NSCI.0123
SOM.MK.I.BPM2.4.NB.4.NSCI.0532
SOM.MK.I.BPM2.1.NB.1.NSCI.0124
SOM.CS.I.BPM2.4.NB.4.NSCI.0125
SOM.MK.IV.BPM2.4.NB.4.NSCI.0126
SOM.MK.I.BPM2.1.NB.1.NSCI.0211
SOM.CS.I.BPM2.1.NB.1.NSCI.0548
SOM.MK.I.BPM2.1.NB.1.NSCI.0213

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0136
SOM.MK.I.BPM2.1.NB.1.NSCI.0137
SOM.MK.I.BPM2.1.NB.1.NSCI.0144
SOM.MK.I.BPM2.1.NB.1.NSCI.0145
SOM.MK.I.BPM2.1.NB.1.NSCI.0149
SOM.MK.I.BPM2.1.NB.1.NSCI.0150

Lecture NSCI

Classify afferent fibers according to numerical and alphabetical nomenclature, diameter, conduction velocity and somatosensory modality
Use knowledge of dermatomal landmarks to interpret clinical findings and demonstrate the ability to extrapolate for regions in between
Describe the two pathways of the somatosensory system, the Dorsal Column / Medial Lemniscus System and the Anterolateral System from the dermatomes of the skin to the spinal cord, including their passage within the spinal cord, and the topographical organization of the fibers
Describe the anatomical details of Lissauer’s Tract and explain its relevance in clinical case scenarios
Use knowledge of the anatomical and functional organization of the somatosensory system to identify and localize lesions associated with Brown-Sequard syndrome and Syringomyelia

Syringomyelia

Critically appraise the symptoms and underlying pathology(ies) of Syringomyelia and discuss the differential diagnoses, possible site(s) of the lesion(s) and nature of the disease.
Apply knowledge of spinal landmarks of the motor system and reflexes to syringomyelia.
In the context of syringomyelia, list the deep tendon reflexes for spinal levels C5, C6 (biceps, brachioradialis); C7, C8 (triceps); L2.
In the context of syringomyelia, describe the two pathways of the somatosensory system, the Dorsal Column / Medial Lemniscus System and the Anterolateral System from the dermatomes of the skin to the spinal cord, including their passage within the spinal cord, and the topographical organization of the fibers.

Touch

Describe the complete pathways for touch, vibration and proprioception, including the localization in different sections of the central nervous system
Use knowledge of touch, vibration and proprioception pathways in a clinical scenario to localize a lesion
Identify the primary somatosensory cortex and localize the areas involved anatomically, as well as in Brodmann’s system
Use knowledge of the somatotopic organization of primary somatosensory cortex in a clinical scenario to localize a lesion.
Explain two point discrimination, the variation in different areas of the body’s surface, and its underlying mechanisms
Describe the procedures for the neurologic examination of touch, vibration, proprioception, two point discrimination, stereognosis and graphesthesia , and interpret normal and pathological outcomes
Explain phantom limb sensations using knowledge of functional and anatomical principles of the somatosensory system.

Pain

Describe the different types of pain and explain the underlying mechanisms
Describe the complete pathways for pain and temperature, including the localization in different sections of the central nervous system
Use the knowledge of pain and temperature pathways in a clinical scenario to localize a lesion
Explain the difference between, and the chemical basis of, activation and sensitization of nociceptors
Describe the physiological changes of pain perception taking place after injury and differentiate allodynia from hyperalgesia
Describe the gate control theory, the afferent and descending pain regulation pathways and explain the transmitters and mechanisms involved
Describe the procedures for the neurologic examination of pain, and interpret normal and pathological outcomes
Relate pharmacological and surgical management of pain to underlying functional and anatomical principles
Describe the corneal reflex, its function, circuitry, and the afferent and efferent limbs, including the adequate stimulus and responses, and the innervation of the eye lid
Describe and explain the neurologic examination of the corneal reflex and use it as a means of localizing lesions by relating the findings to the underlying anatomical and functional principles
Explain Bell’s palsy and the deficits it causes in relation to the corneal reflex and facial expression

Introduction to the Motor System

Describe the general organization of the motor system
Contrast lower versus upper motor neurons
Localize the primary motor cortex
Describe the somatotopic organization and the outputs of the primary motor cortex
Describe the origin, route, and termination of the lateral corticospinal pathway
Localize and describe the corticospinal decussation

Pathways in the Motor System

SOM.MK.I.BPM2.1.NB.1.NSCI.0146
SOM.MK.I.BPM2.1.NB.1.NSCI.0148
SOM.MK.IV.BPM2.4 .NB.2.NSCI.0140
SOM.MK.IV.BPM2.4.NB.2.NSCI.0141
SOM.MK.IV.BPM2.4.NB.2.NSCI.0159
SOM.MK.II.BPM2.4.NB.2.NSCI.0152
SOM.MK.I.BPM2.4.NB.2.NSCI.0153

Small group NSCI
SOM.CS.IV.BPM2.5.NB.5.NSCI.0375
SOM.CS.III.BPM2.5.NB.5.NSCI.0376
SOM.CS.IV.BPM2.5.NB.5.NSCI.0377
SOM.CS.IV.BPM2.5.NB.5.NSCI.0577
SOM.CS.IV.BPM2.5.NB.5.NSCI.0378
SOM.CS.IV.BPM2.5.NB.5.NSCI.0379
SOM.CS.IV.BPM2.5.NB.5.NSCI.0380
SOM.CS.IV.BPM2.5.NB.5.NSCI.0578
SOM.CS.III.BPM2.5.NB.5.NSCI.0381
SOM.CS.III.BPM2.5.NB.5.NSCI.0382
SOM.CS.IV.BPM2.5.NB.5.NSCI.0383
SOM.MK.II.BPM2.5.NB.5.NSCI.0384

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0160
SOM.MK.I.BPM2.1.NB.1.NSCI.0161
SOM.MK.I.BPM2.1.NB.2.NSCI.0162
SOM.MK.I.BPM2.1.NB.1.NSCI.0163
SOM.MK.II.BPM2.1.NB.1.NSCI.0139
SOM.MK.IV.BPM2.4.NB.2.NSCI.0164
SOM.MK.I.BPM2.1.NB.1.NSCI.0166
SOM.MK.I.BPM2.1.NB.1.NSCI.0167

Lecture NSCI
SOM.MK.II.BPM2.4.NB.2.NSCI.0171
SOM.MK.II.BPM2.4.NB.2.NSCI.0172
SOM.MK.II.BPM2.4.NB.2.NSCI.0173
SOM.MK.II.BPM2.4.NB.2.NSCI.0174
SOM.MK.II.BPM2.4.NB.2.NSCI.0175
SOM.MK.II.BPM2.4.NB.2.NSCI.0176
SOM.MK.II.BPM2.4.NB.2.NSCI.0177
SOM.MK.II.BPM2.4.NB.2.NSCI.0178

Lecture NSCI
SOM.MK.I.BPM2.4.NB.2.NSCI.0180
SOM.MK.I.BPM2.1.NB.1.NSCI.0181

Describe cortical inputs to the primary motor area
Describe interactions of premotor and primary motor cortices
Define the scale for assessment of reflexive responses
Explain the normal and abnormal plantar reflex
Explain the neurological bases of decorticate and decerebrate posturing
Contrast common manifestations of lower versus upper motor neuron lesions
Correlate upper and lower motor neuronal signs with lesions affecting central and peripheral nervous sites

NB SG 07 - Neuro Exam 01 (Sensory-Motor)

Assess painful sensation using a gentle pinprick over different regions of the body: fingers, arms, torso (including back), legs. Note any differences between the responses
Appraise the procedure to test warm and cold sensation, including the apparatus used and the regions tested
Measure two-point discrimination in a volunteer's hand (including the ulnar and median nerve distributions), calf, shin, front and back of the torso.
Rationalize the quantitative differences between the two-point discrimination measured in various territories
Examine the steps involved in the testing of proprioception.
Test vibratory sensation at the bony prominences using the 256 Hz tuning fork
Assess muscle strength, bulk and tone
Discuss the grading and significance of abnormal responses
Associate elicited deep tendon reflexes for spinal levels C5, C6 (biceps, brachioradialis); C7, C8 (triceps); L2-L4 (patellar); S1 (Achilles) with the different levels of reflex response.
Examine testing of the abdominal reflexes for spinal levels T8-T10; T10-T12.
Assess normal plantar response and the significance of an abnormal plantar response.
Analyze the clinical presentation of unilateral spinal lesions.

Muscle Innervation and Spinal Reflexes

Describe the physiological role of the muscle spindle
Describe the physiological role of the Golgi tendon organ
Describe opposition of the myotatic reflex by the inverse myotatic reflex
Explain the flexion and flexion crossed extension reflexes
Identify major spinal landmarks of the motor system and reflexes
Explain neurological bases of areflexia/hyporeflexia
Define the motor unit and characterize the types of motor units within the muscle
Describe the mechanisms whereby contraction of a muscle can be increased

Diseases of the NMJ and Motor Unit

Identify pathological sites within the motor unit leading to muscle weakness
Identify diseases affecting the motor unit leading to weakness
Characterize lesions of the cell bodies or axons forming part of the motor unit
Describe demyelinating lesions affecting the motor unit
Describe lesions of the axonal terminals, synaptic cleft and end-plate within the motor unit
Describe lesions of somatic muscle
Describe denervation and reinnervation in relation to lower motor neuron syndrome
Correlate clinical signs of disease with lesions of the motor unit

PNS Disorders

Define negative and positive manifestations of peripheral nervous disease
Explain the concept and measurement of nerve conduction velocity

SOM.MK.II.BPM2.4.NB.2.NSCI.0182
SOM.MK.II.BPM2.4.NB.2.NSCI.0183
SOM.MK.II.BPM2.4.NB.2.NSCI.0184
SOM.MK.II.BPM2.4.NB.2.NSCI.0185

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0430
SOM.CS.II.BPM2.4.NB.4.NSCI.0431
SOM.CS.II.BPM2.4.NB.4.NSCI.0432
SOM.CS.III.BPM2.4.NB.4.NSCI.0433
SOM.CS.II.BPM2.4.NB.4.NSCI.0434
SOM.CS.III.BPM2.4.NB.4.NSCI.0435
SOM.CS.II.BPM2.4.NB.4.NSCI.0436
SOM.CS.II.BPM2.4.NB.4.NSCI.0437
SOM.CS.II.BPM2.4.NB.4.NSCI.0438
SOM.CS.III.BPM2.4.NB.4.NSCI.0439
SOM.CS.II.BPM2.4.NB.4.NSCI.0440
SOM.CS.II.BPM2.4.NB.4.NSCI.0441
SOM.CS.III.BPM2.4.NB.4.NSCI.0442

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0187
SOM.MK.I.BPM2.1.NB.1.NSCI.0188
SOM.MK.I.BPM2.1.NB.1.NSCI.0189
SOM.MK.II.BPM2.4.NB.2.NSCI.0190
SOM.MK.II.BPM2.4.NB.2.NSCI.0191
SOM.MK.II.BPM2.4.NB.2.NSCI.0192
SOM.MK.IV.BPM2.4.NB.4.NSCI.0193

Small group NSCI
SOM.CS.III.BPM2.5.NB.5.NSCI.0385
SOM.CS.IV.BPM2.5.NB.5.NSCI.0386
SOM.CS.IIII.BPM2.5.NB.5.NSCI.0387
SOM.CS.IV.BPM2.5.NB.5.NSCI.0388
SOM.CS.IV.BPM2.5.NB.5.NSCI.0389
SOM.CS.IV.BPM2.5.NB.5.NSCI.0390
SOM.CS.IV.BPM2.5.NB.5.NSCI.0391
SOM.CS.IV.BPM2.5.NB.5.NSCI.0579
SOM.CS.IV.BPM2.5.NB.5.NSCI.0392
SOM.CS.IV.BPM2.5.NB.5.NSCI.0393
SOM.MK.I.BPM2.5.NB.5.NSCI.0394

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0197
SOM.MK.I. BPM2.4.NB.2.NSCI.0198

Describe the structure of a peripheral nerve and traumatic events leading to structural damage and dysfunction (i.e., compression, crush, axotomy)
Explain events related to muscular denervation and reinnervation following lower motor neuronal axotomy
Explain acute demyelinating neuropathy (e.g., Guillain-Barre syndrome)
Explain chronic forms of neuropathy (e.g., lead poisoning, alcoholic, diabetes mellitus)

Guillain-Barre Syndrome

In the context of elevated intracranial pressure, describe upper and lower motor neuron signs.
Classify lesions in Guillain-Barre syndrome as peripheral or central.
In the context of GBS, describe the complete pathways for touch, vibration and proprioception, including the localization in different sections of the central nervous system
Apply knowledge of pathways for touch, vibration and proprioception pathways in Guillain-Barre syndrome.
In the context of GBS, describe the complete pathways for pain and temperature, including the localization in different sections of the central nervous system.
Apply knowledge of pain and temperature pathways in Guillai-Barre syndrome.
In the context of GBS, describe common manifestations of lower motor neuron lesions.
Apply knowledge of the lateral corticospinal pathway to Guillain-Barre disease.
In the context of GBS, explain the concept and measurement of nerve conduction velocity.
Apply assessment of nerve conduction velocity to Guillain-Barre syndrome.
Apply knowledge of upper and lower motor neurons signs in Guillain-Barre syndrome.
In the context of GBS, describe movement of solute (and water) across the BBB and blood.
Describe laboratory findings in the context of Guillain-Barre syndrome.

Basal Ganglia

Identify major components of the basal ganglia and general principles governing their functional interactions
Describe the major synaptic connections that form the direct and the indirect basal gangliar motor loops
Describe the influence of basal gangliar components on cortical activity and hence movement
Relate the nature and causes of Parkinsonism to the direct and indirect motor loops
Relate the nature and causes of hyperkinesia to the direct and indirect motor loops
Relate therapeutic approaches to Parkinson disease to changes in activity within the direct and indirect basal gangliar motor loops
Relate therapeutic approaches to Huntington disease to changes in activity within the direct and indirect basal gangliar motor loops

NB SG 08 - Neuro Exam 02 (Cranial Nerve Testing I, II, III, IV, VI)

Demonstrate the functional assessment of CNI
Demonstrate operation of the Snellen Eye Chart/Pocket Eye Card to test visual acuity
Examine the terms counting fingers (CF), hand motion (HM), light perception (LP), no light perception (NLP)
Demonstrate testing of color vision.
Demonstrate confrontation visual field testing.
Demonstrate Mapping of the Blind Spot
Demonstrate testing of extraocular muscle function (H-test)
Interpret pathological findings at each cardinalpoint of gaze with respect to the underlying CN involvemnet
Demonstrate testing of the pupillary light reflex.
Demonstrate the use of the ophthalmoscope in visualizing the optic fundi, retina and associated vessels.
Analyze the clinical presentation of a compressive diencephalic lesion.

Cerebellum

Describe the normal functions of the cerebellum
Articulate basic rules governing cerebellar function and dysfunction

SOM.MK.I.BPM2.1.NB.1.NSCI.0199
SOM.MK.I.BPM2.1.NB.1.NSCI.0200
SOM.MK.I.BPM2.1.NB.1.NSCI.0201
SOM.MK.I.BPM2.4.NB.2.NSCI.0202
SOM.MK.I.BPM2.4.NB.2.NSCI.0203

Describe regional variation in cerebellar function and significance of cerebellar peduncles
Describe major connections between the cerebellum and the motor systems
Describe origins and routes of cerebellar afferent pathways
Correlate cerebellar lesion sites and clinical manifestations
Describe the clinical manifestations of select cerebellar diseases (cerebellar tumor, cerebellar stroke, alcohol-induced thiamine deficiency and Louis-Bar Syndrome)

DLA NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0195
SOM.MK.I.BPM2.1.NB.1.NSCI.0196

Cerebellum
Describe cellular and synaptic organization of the cerebellar cortex
Correlate cerebellar cortical zones and deep cerebellar nuclei

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0256
SOM.MK.I.BPM2.1.NB.1.NSCI.0257
SOM.MK.I.BPM2.1.NB.1.NSCI.0258
SOM.MK.I.BPM2.4.NB.2.NSCI.0259
SOM.MK.I.BPM2.4.NB.2.NSCI.0260
SOM.MK.I.BPM2.4.NB.2.NSCI.0261

Flipped Class: Integrative Neuroimaging
Integrate knowledge of CT, T1-weighted MRI and T2-weighted MRI neuroimages
Integrate knowledge of modality, projection, phase and site of contrast application in angiographic images
Integrate knowledge of neuroanatomy and neuroimaging
Integrate knowledge of common manifestations of upper motor neuron lesions.
Integrate knowledge of common manifestations of lower motor neuron lesions.
Integrate knowledge of upper and lower motor neuronal signs with lesions affecting central and peripheral nervous sites.

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0206
SOM.MK.I.BPM2.1.NB.1.NSCI.0207
SOM.MK.I.BPM2.4.NB.4.NSCI.0208
SOM.MK.I.BPM2.1.NB.1.NSCI.0209
SOM.MK.I.BPM2.4.NB.4.NSCI.0214
SOM.MK.I.BPM2.4.NB.4.NSCI.0215
SOM.MK.I.BPM2.4.NB.4.NSCI.0216
SOM.MK.I.BPM2.1.NB.1.NSCI.0217
SOM.MK.I.BPM2.1.NB.1.NSCI.0218
SOM.CS.I.BPM2.4.NB.4.NSCI.0219
SOM.MK.II.BPM2.4.NB.4.NSCI.0220
SOM.MK.I.BPM2.1.NB.1.NSCI.0205

Visual System: Retina
Review the major structures of the eye and where they are located, their functional relevance, including the blind spot, and interpret the fundus of the eye as it is seen through an ophthalmoscope
Explain how refraction contributes to the formation of an image, recognize the refractive power of cornea and lens
Explain the mechanism of near and far vision, define refractive power for the different structures of the eye, and describe the change of refractive plasticity during aging
Define visual acuity and explain the major factors contributing to visual acuity
Describe the procedures for the neurologic examination of Visual Acuity and the Examination of the Ocular Fundus (Fundoscopy), and interpret normal and pathological outcomes
Contrast features of five ocular conditions leading to visual impairment
Interpret the fundus of the eye as it is seen through an ophthalmoscope and relate the optic nerve head and the macula to the blind spot and central vision
Describe the structure of retinal photoreceptors and their functional properties
Describe the functional characteristics of the three different cone types and explain how they contribute to color vision
Describe the testing procedure for Color Blindness , interpret normal and pathological outcomes, and explain the different frequencies in males and females based on the underlying genetics
Relate the clinical correlations of Retinitis Pigmentosa, Night Blindness, and Color Blindness to the underlying functional and anatomical principles
Describe reflexive control of pupillary diameter

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0221
SOM.MK.I.BPM2.1.NB.1.NSCI.0222
SOM.MK.I.BPM2.1.NB.1.NSCI.0223
SOM.MK.I.BPM2.1.NB.1.NSCI.0210
SOM.CS.I.BPM2.1.NB.1.NSCI.0549
SOM.MK.I.BPM2.1.NB.1.NSCI.0224
SOM.MK.I.BPM2.1.NB.1.NSCI.0225
SOM.MK.I.BPM2.1.NB.1.NSCI.0226
SOM.MK.I.BPM2.1.NB.4.NSCI.0228

Visual Pathways
Describe the organization of the visual fields including the anatomical correlates of the point of fixation and the blind spot
Describe how the visual field is represented on the retina
Describe the retinal projections to the different areas of the brain
Describe and explain the Pupillary Light Reflex, its function, circuitry, and the afferent and efferent limbs, including the adequate stimulus and responses
Describe and explain the neurologic examination of the pupillary light reflex and use it as a means of localizing lesions by relating the findings to the underlying anatomical and functional principles
Illustrate and explain the neuronal pathways for all four quadrants of the visual field
Identify and describe the location of the primary visual cortex in the occipital lobe
Describe the organization of parallel functional pathways for depth, motion, form, color, word form and face processing
Relate the clinical correlations of Visual Pathway Lesions, with and without Sparing of the Macula, and including Achromatopsia, alexia and prosopognosia to the underlying functional and anatomical principles

Small group NSCI
SOM.CS.IV.BPM2.5.NB.5.NSCI.0395

NB SG 09 - Neuro Exam 03 (Cranial Nerve Testing V, VII, VIII, IX, X, XI, XII)
Demonstrate testing of the corneal reflex and interpret normal and all possible pathological findings with respect to the underlying pathway.

SOM.CS.III.BPM2.5.NB.5.NSCI.0396
SOM.CS.III.BPM2.5.NB.5.NSCI.0397
SOM.CS.II.BPM2.5.NB.5.NSCI.0398
SOM.CS.IV.BPM2.5.NB.5.NSCI.0399

SOM.CS.IV.BPM2.5.NB.5.NSCI.0400
SOM.CS.IV.BPM2.5.NB.5.NSCI.0401
SOM.CS.III.BPM2.5.NB.5.NSCI.0402
SOM.CS.IV.BPM2.5.NB.5.NSCI.0403
SOM.CS.IV.BPM2.5.NB.5.NSCI.0404
SOM.CS.IV.BPM2.5.NB.5.NSCI.0405
SOM.MK.I.BPM2.5.NB.5.NSCI.0406

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0231
SOM.MK.I.BPM2.1.NB.1.NSCI.0232
SOM.MK.I.BPM2.1.NB.1.NSCI.0233
SOM.CS.I.BPM2.1.NB.4.NSCI.0234

SOM.MK.I.BPM2.1.NB.4.NSCI.0235

DLA NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0229
SOM.MK.I.BPM2.1.NB.1.NSCI.0230

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0236
SOM.MK.I.BPM2.1.NB.1.NSCI.0237
SOM.MK.I.BPM2.1.NB.1.NSCI.0238
SOM.MK.I.BPM2.1.NB.1.NSCI.0239
SOM.MK.I.BPM2.1.NB.4.NSCI.0240
SOM.MK.I.BPM2.1.NB.1.NSCI.0241
SOM.MK.I.BPM2.1.NB.1.NSCI.0242

SOM.MK.I.BPM2.1.NB.1.NSCI.0243
SOM.CS.I.BPM2.1.NB.4.NSCI.0244
SOM.MK.I.BPM2.1.NB.4.NSCI.0245

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0250
SOM.MK.I.BPM2.1.NB.1.NSCI.0251
SOM.CS.I.BPM2.1.NB.4.NSCI.0252
SOM.MK.II.BPM2.1.NB.4.NSCI.0253
SOM.MK.I.BPM2.1.NB.1.NSCI.0254
SOM.MK.I.BPM2.1.NB.1.NSCI.0255

Assess the motor portion of CN V by testing the masseter muscles and jaw-jerk reflex. Distinguish between normal and pathological findings.
Illustrate testing CN VII sensory function.

Contrast upper and lower motor neuron signs in testing of the motor component of cranial nerve VII.
Demonstrate use of the otoscope to inspect the outer ear and identify the visible structures.
Demonstrate use of the 512Hz tuning fork during assessment of auditory function – perform both Weber’s and Rinne’s tests and distinguish between sensorineural and conductive hearing loss. Discriminate between normal and pathological findings.
Demonstrate the induction of optokinetic nystagmus using the OKN tape.
Assess the functional testing of CN VIII in a comatose individual.
Demonstrate examination of palatal elevation to test CN IX and CN X. Describe other assessments of these cranial nerves.
Assess functional testing of CN XI (sternocleidomastoid and trapezoid muscles) in a patient and rationalize any possible pathological findings
Demonstrate the clinical assessment of cranial nerve XII.
Analyze the clinical findings associated with a compressive brainstem tumor.

Eye Movements

Localize the ocular motor nuclei and other control units of extraocular movements in the brain stem and in the cerebral cortex
Describe the function and characteristic parameters of saccadic eye movements and lateral gaze
Describe the neuronal circuitry of saccades
Describe the neurologic examination of eye movements and interpret normal and pathological outcomes
Relate the clinical correlations of Trochlear Nerve Palsy, Oculomotor Nerve Palsy, Abducens Nerve Palsy, Diplopia, Internuclear Ophthalmoplegia, PPRF Lesion and One-and-a-half Syndrome to the underlying functional and anatomical principles

Classification of Eye Movements, Eye Simulator, Extraocular Muscles

Describe eye movements
Name the extraocular muscles, list their innervation, and the primary function of each muscle, and interpret graphical sketches of extraocular function

Auditory System

Explain the mechanisms of pressure amplification in the middle ear
Describe the function of the tensor tympani and stapedius muscles and explain their role in the attenuation reflex
Explain the mechanism for the encoding of sound frequency
Explain the mechanism of sound amplification in the inner ear
List the elements of the auditory pathways in the peripheral nervous system and the different levels of the central nervous system
Localize the primary auditory cortex
Describe the organization of the primary auditory cortex, including its tonotopic map

Relate the manifestations of conductive versus sensorineural hearing loss, unilateral versus bilateral hearing loss, acquired versus age-related hearing loss and Meniere’s disease to the underlying functional and anatomical alterations
Describe clinical assessment of auditory function
Describe presbycusis and explain the basis of its associated frequency specific auditory deficits

Ocular Reflexes

Explain the mechanism and circuitry of the afferent and efferent limbs of the horizontal vestibulo-ocular reflex
Describe mechanisms of nystagmus
Explain the use of caloric testing and oculocephalic maneuver to examine brainstem function in a comatose patient
Correlate Motion Sickness, Alcohol Intoxication, Antibiotics, and the Pathological Vestibular Nystagmus with underlying functional and anatomical principles
Explain the induction, mechanism and circuitry of the afferent and efferent limbs of the optokinetic reflex
Describe mechanisms underlying optokinetic nystagmus

DLA NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0246
SOM.MK.I.BPM2.1.NB.1.NSCI.0247
SOM.MK.I.BPM2.1.NB.1.NSCI.0248
SOM.MK.I.BPM2.1.NB.1.NSCI.0249
SOM.CS.I.BPM2.1.NB.4.NSCI.0533

Lecture NSCI
SOM.MK.I.BPM2.1.NB.1.NSCI.0274
SOM.MK.I.BPM2.1.NB.1.NSCI.0275
SOM.MK.I.BPM2.1.NB.1.NSCI.0276
SOM.MK.I.BPM2.1.NB.1.NSCI.0277
SOM.MK.I.BPM2.1.NB.1.NSCI.0278
SOM.MK.I.BPM2.1.NB.1.NSCI.0279
SOM.MK.I.BPM2.4.NB.2.NSCI.0280

DLA NSCI
SOM.MK.1.BPM2.1.NB.1.NSCI.0281
SOM.MK.I.BPM2.1.NB.1.NSCI.0282

Lecture NSCI
SOM.MK.I.BPM2.1.NB.4.NSCI.0534
SOM.CS.II.BPM2.4.NB.4.NSCI.0535
SOM.CS.II.BPM2.4.NB.4.NSCI.0536
SOM.CS.II.BPM2.4.NB.4.NSCI.0537
SOM.CS.II.BPM2.4.NB.4.NSCI.0538
SOM.CS.II.BPM2.4.NB.4.NSCI.0539
SOM.CS.II.BPM2.4.NB.4.NSCI.0540
SOM.CS.II.BPM2.4.NB.4.NSCI.0541

Lecture BEHS
SOM.MK.III.BPM2.4.NB.3.BEHS.0003
SOM.MK.III.BPM2.4.NB.3.BEHS.0004
SOM.MK.III.BPM2.4.NB.3.BEHS.0005
SOM.MK.III.BPM2.4.NB.3.BEHS.0006
SOM.MK.III.BPM2.4.NB.3.BEHS.0007
SOM.MK.III.BPM2.4.NB.3.BEHS.0008
SOM.MK.III.BPM2.4.NB.3.BEHS.0009
SOM.MK.III.BPM2.4.NB.3.BEHS.0155

DLA BEHS
SOM.MK.III.BPM2.4.NB.3.BEHS.0001
SOM.MK.III.BPM2.4.NB.3.BEHS.0002

Neurologic Examination of the Optokinetic Nystagmus, Vestibilar System

Describe the functional anatomy of the two receptor organs of the vestibular system, the otolithic organs and the semicircular canals
Explain the cellular and molecular elements of the signal transduction process in the inner ear
Explain how linear acceleration is detected in utricle and saccule, and angular acceleration in the semicircular canals
Describe the vestibular pathways
Describe the Neurologic Examination of the Optokinetic Nystagmus

Autonomic Nervous System

Describe regulation of the autonomic nervous system by circuitry within the brain.
Describe hypothalamic function as the central control unit of life, including the integration of autonomic function, endocrine function and behavior
Describe the neuroendocrine function in the context of chemical signaling in general and compare it with other modes of chemical signaling
Describe the structure of the hypothalamo-pituitary axis.
Describe the functional connectivity between the hypothalamus and anterior and posterior pituitary.
Summarize the control circuits of the urinary bladder, including the different levels of bladder control, and the control mechanisms and circuits of its filling and emptying
Contrast automatic and atonic bladder.

Autonomic Nervous System

Describe hypothalamic control of feeding, water balance and body temperature.
Outline the innervation and autonomic control of reproductive organs

Flipped Class: Principles of Lesion Localization

Discuss general principles of lesion localization, based on functional assessment of nerves
Identify telencephalic lesions by assessment of cranial nerves
Identify diencephalic lesions by assessment of cranial nerves
Identify mesencephalic lesions by assessment of cranial nerves
Identify metencephalic lesions by assessment of cranial nerves
Identify myelencephalic lesions by assessment of cranial nerves
Identify spinal lesions by assessment of spinal nerves
Identify telencephalic lesions by assessment of cranial nerves

Lifespan Development I: Age 0 - 2 years

Describe the major domains of human development (physical, cognitive, and socio-emotional)
Describe the primitive reflexes and capabilities of the newborn
Describe the major physical, cognitive, and socio-emotional milestones during the first year of life
Describe the major physical, cognitive, and socio-emotional milestones during the second year of life
Describe Piaget’s four stages of cognitive development
Identify appropriate assessment methods to screen for delays in developmental milestones during the first two years of life
Identify developmental red flags during the first two years of life
Describe age-appropriate nutritional recommendations during infancy, childhood, and adolescence

Emotional Development of the Child

Describe how temperament, early childhood attachment, and parent-child relational challenges influence development
Explain the long-term effects of infant deprivation, child abuse, child neglect, and vulnerable child syndrome on development

DLA BEHS SOM.MK.III.BPM2.4.NB.3.BEHS.0010	Erikson’s Stages of Psychosocial Development Describe Erikson’s stages of psychosocial development
Small group NSCI SOM.MK.I.BPM2.5.NB.5.NSCI.0571	NB SG 10 - Section Anatomy of the CNS with Clinical Cases Assess the location of key structures of the central nervous system, using magnetic resonance images, and well as the functional relevance of these structures
SOM.MK.I.BPM2.5.NB.5.NSCI.0572	Apply the knowledge of section anatomy of the central nervous system, pathways and cranial nerve functions to localize the major vascular lesions of the cerebrum, and explain the functional deficits associated with these lesions
SOM.MK.I.BPM2.5.NB.5.NSCI.0573	Apply the knowledge of section anatomy of the central nervous system, pathways and cranial nerve functions to localize the major vascular brainstem lesions, and explain the functional deficits associated with these lesions
SOM.MK.I.BPM2.5.NB.5.NSCI.0574	Apply the knowledge of section anatomy of the central nervous system, pathways and cranial nerve functions to localize the major vascular lesions of the cerebellum, and explain the functional deficits associated with these lesions
Lecture BEHS SOM.MK.III.BPM2.4.NB.3.BEHS.0012 SOM.MK.III.BPM2.4.NB.3.BEHS.0014 SOM.MK.III.BPM2.4.NB.3.BEHS.0011 SOM.MK.III.BPM2.4.NB.3.BEHS.0007 SOM.MK.III.BPM2.4.NB.3.BEHS.0016 SOM.MK.III.BPM2.4.NB.3.BEHS.0018 SOM.MK.III.BPM2.4.NB.3.BEHS.0155	Lifespan Development II: Age 3 years - Adolescence Describe the major physical, cognitive, and socio-emotional milestones throughout childhood Describe the major physical, cognitive, and socio-emotional developmental trends throughout adolescence Describe general trends in brain development during childhood and adolescence Describe Piaget’s four stages of cognitive development Describe how adverse childhood experiences (ACEs), such as poverty, abuse, or neglect, can affect development Discuss the role of adolescent brain maturation, particularly the prefrontal cortex and limbic system, in risk-taking behaviors Describe age-appropriate nutritional recommendations during infancy, childhood, and adolescence
Lecture BEHS SOM.MK.III.BPM2.4.NB.3.BEHS.0019 SOM.MK.III.BPM2.4.NB.3.BEHS.0022 SOM.MK.III.BPM2.4.NB.3.BEHS.0024 SOM.MK.IV.BPM2.4.NB.3.BEHS.0025	Lifespan Development III: Adulthood - End of Life Describe the major physical, cognitive, and socio-emotional trends during adulthood Discuss predominant health issues and risks across the lifespan Apply Kübler-Ross' five stages of dying theory by illustrating each stage with examples of patient experiences Describe the principles of palliative care and hospice care, highlighting their key differences and shared goals
Lecture BEHS SOM.PB.IV.BPM2.5.NB.5.BEHS.0026 SOM.CS.I.BPM2.5.NB.5.BEHS.0027 SOM.CS.I.BPM2.5.NB.5.BEHS.0028 SOM.CS.I.BPM2.5.NB.5.BEHS.0029 SOM.MK.III.BPM2.5.NB.5.BEHS.0030	Physician-Patient Relationship Describe the rights and responsibilities of the physician and the patient Discuss the importance of effective communication for medical diagnosis, patient satisfaction, and treatment outcomes Describe barriers to effective communication and strategies to overcome them Identify factors that decrease treatment adherence and how best to address them Describe the “sick role” and how it might negatively impact recovery from illness
Lecture BEHS SOM.CS.III.BPM2.5.NB.5.BEHS.0031 SOM.CS.III.BPM2.5.NB.5.BEHS.0032 SOM.CS.III.BPM2.5.NB.5.BEHS.0033 SOM.CS.I.BPM2.5.NB.5.BEHS.0034 SOM.CS.II.BPM2.5.NB.5.BEHS.0035 SOM.CS.III.BPM2.5.NB.5.BEHS.0036 SOM.CS.I.BPM2.5.NB.5.BEHS.0037 SOM.CS.I.BPM2.5.NB.5.BEHS.0038	Patient-Centered Interviewing Describe data gathering techniques used during the clinical interview Identify the key components of the medical encounter Identify the basic types of information gathered during a patient-physician encounter Explain how empathy strengthens the patient-physician relationship and contributes to a more effective partnership Identify the basic components of good clinical decision-making Describe the types of information documented under each section of a SOAP note during the patient-physician interaction Describe the main principles of motivational interviewing Assess the patient’s stage of behavior change and use motivational interviewing to tailor the approach for effective health behavior modification
Lecture BEHS	Psychiatric Interviewing and Mental Status Examination

SOM.CS.III.BPM2.5.NB.5.BEHS.0039	Describe differences between medical and psychiatric interviews
SOM.CS.III.BPM2.5.NB.5.BEHS.0040	Describe the major components of a mental status examination
SOM.CS.III.BPM2.5.NB.5.BEHS.0041	Use and differentiate terminology for abnormalities across all mental status domains (e.g., appearance, behavior, speech/language, mood/affect, thought, perception, insight, judgment, cognition)
SOM.CS.III.BPM2.5.NB.5.BEHS.0042	Describe how to conduct a clinical risk assessment
Lecture BEHS	
SOM.MK.III.BPM2.4.NB.3.BEHS.0049	Flipped Class: Learning Theory
SOM.MK.III.BPM2.4.NB.3.BEHS.0050	Use learning theory terminology to explain the learning and extinction of classically-conditioned responses
SOM.MK.III.BPM2.4.NB.3.BEHS.0051	Use learning theory terminology to explain the learning and extinction of operantly-conditioned responses
	Compare and contrast classical and operant conditioning principles
DLA BEHS	
SOM.MK.III.BPM2.4.NB.3.BEHS.0043	Basics of Classical and Operant Conditioning
SOM.MK.III.BPM2.4.NB.3.BEHS.0045	Explain the basic principles of classical conditioning, including its components, factors influencing its effectiveness, the extinction process, spontaneous recovery, and re-extinction
SOM.MK.III.BPM2.4.NB.3.BEHS.0046	Explain the operant principles of reinforcement, punishment, extinction, extinction burst, and shaping
SOM.MK.III.BPM2.4.NB.3.BEHS.0044	Identify the factors that influence the effectiveness of reinforcers and punishers
SOM.MK.III.BPM2.4.NB.3.BEHS.0047	Describe the disadvantages of using punishment, including the development of learned helplessness
SOM.MK.III.BPM2.4.NB.3.BEHS.0048	Identify the different types of reinforcement schedules used to modify behavior, and explain the advantages and disadvantages of each type
	Explain the role of classical and operant conditioning in psychoactive drug use and treatment, including the concepts of conditioned euphoria and conditioned tolerance
Small group BEHS	
SOM.PB.VI.BPM2.5.NB.5.BEHS.0162	NB SG 11 - Sociocultural Sensitivity and Awareness in Healthcare
	Discuss how social and cross-cultural factors (e.g., language, decision-making styles, family dynamics, beliefs, and practices) influence patient–physician interactions and health care outcomes, emphasizing the role of cultural competence in patient-centered care
Lecture BEHS	
SOM.MK.IV.BPM2.4.NB.3.BEHS.0052	Psychotherapies
SOM.MK.IV.BPM2.4.NB.3.BEHS.0056	Describe the types of behavioral therapies (systematic desensitization, graded exposure, flooding, exposure and response prevention, aversion therapy, biofeedback, token economies, behavioral modeling, behavioral activation), including their components and underlying learning principles
SOM.MK.IV.BPM2.4.NB.4.BEHS.0063	Describe the components of cognitive therapy, including its underlying learning principles
SOM.MK.III.BPM2.4.NB.4.BEHS.0059	Describe the components of cognitive behavioral therapy and its role in pain management
SOM.MK.IV.BPM2.4.NB.4.BEHS.0061	Describe the components of psychodynamic therapy, including its underlying learning principles
	Describe the components of family therapy, including its underlying learning principles
DLA BEHS	
SOM.MK.III.BPM2.4.NB.3.BEHS.0064	Defense Mechanisms
	Identify the defense mechanisms illustrated in a clinical scenario
Lecture NSCI	
SOM.CS.II.BPM2.4.NB.4.NSCI.0474	Flipped Class: Clinical Cases
SOM.CS.II.BPM2.4.NB.4.NSCI.0421	In the context of herniated disc, correlate upper and lower motor neuronal signs with lesions affecting central and peripheral nervous sites.
SOM.CS.II.BPM2.4.NB.4.NSCI.0488	In the context of thalamic infarction, describe the complete pathways for pain and temperature, including the localization in different sections of the central nervous system
SOM.CS.II.BPM2.4.NB.4.NSCI.0407	Critically appraise the symptoms and underlying pathology(ies) of AICA Syndrome, discuss the differential diagnoses, possible site(s) of the lesion(s) and nature of the disease.
SOM.CS.IV.BPM2.4.NB.4.NSCI.0483	Critically appraise the symptoms and underlying pathology(ies) of cerebellar Vestibular schwannoma.
SOM.CS.II.BPM2.4.NB.4.NSCI.0447	In the context of multiple sclerosis, perform confrontation visual field testing.
SOM.CS.II.BPM2.4.NB.4.NSCI.0411	In the context of Wallenberg’s Syndrome, describe the complete pathways for pain and temperature, including the localization in different sections of the central nervous system
SOM.CS.II.BPM2.4.NB.4.NSCI.0448	Critically appraise the symptoms and underlying pathology(ies) of Weber’s syndrome
SOM.CS.II.BPM2.4.NB.4.NSCI.0462	In the context of Wallenberg’s Syndrome, describe the lateral corticospinal pathway from its origins through its termination
SOM.CS.II.BPM2.4.NB.4.NSCI.0443	In the context of Huntington’s disease, localize and functionally characterize the major structures of the diencephalon.
	Critically appraise the symptoms and underlying pathology(ies) of Meniere’s Disease.

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0443
SOM.CS.III.BPM2.4.NB.4.NSCI.0444

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0411
SOM.CS.III.BPM2.4.NB.4.NSCI.0412
SOM.CS.IV.BPM2.4.NB.4.NSCI.0413
SOM.CS.III.BPM2.4.NB.4.NSCI.0414
SOM.CS.IV.BPM2.4.NB.4.NSCI.0415
SOM.CS.II.BPM2.4.NB.4.NSCI.0416
SOM.CS.II.BPM2.4.NB.4.NSCI.0417

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0290
SOM.MK.I.BPM2.4.NB.1.NSCI.0291
SOM.MK.I.BPM2.4.NB.1.NSCI.0292
SOM.MK.I.BPM2.4.NB.1.NSCI.0293
SOM.MK.I.BPM2.4.NB.1.NSCI.0294

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0316
SOM.MK.I.BPM2.4.NB.1.NSCI.0317
SOM.MK.I.BPM2.4.NB.1.NSCI.0582
SOM.MK.I.BPM2.4.NB.1.NSCI.0318
SOM.MK.I.BPM2.4.NB.1.NSCI.0319

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0065
SOM.MK.I.BPM2.4.NB.4.BEHS.0066
SOM.MK.I.BPM2.4.NB.4.BEHS.0067
SOM.MK.I.BPM2.4.NB.4.BEHS.0188
SOM.MK.III.BPM2.4.NB.2.BEHS.0068
SOM.MK.IV.BPM2.4.NB.4.BEHS.0069

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0070
SOM.MK.III.BPM2.4.NB.2.BEHS.0071
SOM.MK.IV.BPM2.4.NB.4.BEHS.0072
SOM.MK.I.BPM2.4.NB.4.BEHS.0177
SOM.MK.I.BPM2.4.NB.4.BEHS.0074
SOM.MK.I.BPM2.4.NB.4.BEHS.0178
SOM.MK.III.BPM2.4.NB.2.BEHS.0077
SOM.MK.IV.BPM2.4.NB.4.BEHS.0078

Meniere's Disease

Critically appraise the symptoms and underlying pathology(ies) of Meniere’s Disease.
In the context of Meniere's Disease, describe and explain the Vestibulo-Ocular Reflex with reference to the horizontal semicircular canals.

Weber's Syndrome

Critically appraise the symptoms and underlying pathology(ies) of Weber's syndrome.
In the context of Weber's syndrome, describe assessment of visual acuity.
In the context of Weber's syndrome, demonstrate testing of extraocular muscle function (H-test).
In the context of Weber's syndrome, describe and explain the Pupillary Light Reflex, its function, circuitry and typical stimuli and responses.
In the context of Weber's syndrome, demonstrate testing of the motor component of CN VII.
In the context of Weber's syndrome, differentiate upper and lower motor neuronal signs.
In the context of Weber's syndrome, trace the lateral corticospinal pathway from its origins through its termination.

Emotions

Describe the basic components of emotion processing according to the James-Lange Theory, the Cannon-Bard Theory, and the Schachter-Singer Two Factor Theory
Outline the history of the Limbic System, from Broca's Limbic Lobe and the Papez Circuit to MacLean's Limbic System and LeDoux’s “High-Road/Low-Road” System
Explain how lesions can help to identify brain structures involved in emotions by describing Klüver-Bucy Syndrome, the case of Phineas Gage, and frontal lobotomies
Understand how dysfunction in the neural circuitry for fear learning, extinction learning, and contextual fear learning can contribute to psychopathology
Relate common symptoms of temporal lobe epilepsy to dysfunction in brain networks underlying emotional expression and emotion regulation

Emotions

Describe evidence for sexual dimorphism of the brain, including frequencies of disease processes, sexual and non-sexual behavior and the hormonal environments of males and females
Describe some of the sexually dimorphic areas of the brain.
Describe the general location and the nature of the differences of the sexually dimorphic areas of the brain
Explain the findings linking morphological differences of brain structures with gender identity and sexual orientation
Discuss sexual differentiation in relation to genetics, hormones, critical periods and the role of testosterone

Introduction to Psychopathology & Anxiety Disorders

Describe the basic features of the DSM-5 classification system
Apply the diagnostic specifiers of provisional, unspecified, prior history, and in remission when diagnosing a patient
Apply the essential diagnostic criteria for anxiety disorders, including any subtypes
Clinically differentiate the anxiety disorders from other conditions
Describe the biopsychosocial factors contributing to anxiety disorders
Describe the management of anxiety disorders, including mechanisms of action and potential side effects

Obsessive-Compulsive and Related Disorders & Somatic Symptom and Related Disorders

Apply the essential diagnostic criteria for obsessive-compulsive and related disorders
Describe the biopsychosocial factors contributing to obsessive-compulsive and related disorders
Describe the management of obsessive-compulsive and related disorders, including mechanisms of action and potential side effects
Clinically differentiate the obsessive-compulsive and related disorders from other conditions
Apply the essential diagnostic criteria for the somatic symptom and related disorders
Clinically differentiate the somatic symptom and related disorders from other conditions, including malingering and malingering by proxy
Describe the biopsychosocial factors contributing to somatic symptom and related disorders
Describe the management of somatic symptom and related disorders, including mechanisms of action and potential side effects

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0079
SOM.MK.I.BPM2.4.NB.4.BEHS.0179
SOM.MK.III.BPM2.4.NB.2.BEHS.0080
SOM.MK.IV.BPM2.4.NB.4.BEHS.0081
SOM.MK.I.BPM2.4.NB.4.BEHS.0082
SOM.MK.I.BPM2.4.NB.4.BEHS.0180
SOM.MK.III.BPM2.4.NB.2.BEHS.0083
SOM.MK.IV.BPM2.4.NB.4.BEHS.0084

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0088
SOM.MK.I.BPM2.4.NB.4.BEHS.0189
SOM.MK.II.BPM2.4.NB.2.BEHS.0086
SOM.MK.IV.BPM2.4.NB.4.BEHS.0087
SOM.MK.I.BPM2.4.NB.4.BEHS.0195
SOM.MK.I.BPM2.4.NB.4.BEHS.0089
SOM.MK.III.BPM2.4.NB.2.BEHS.0090
SOM.MK.IV.BPM2.4.NB.4.BEHS.0091

DLA BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0088
SOM.MK.I.BPM2.4.NB.4.BEHS.0189
SOM.MK.I.BPM2.4.NB.4.BEHS.0092
SOM.MK.I.BPM2.4.NB.4.BEHS.0190
SOM.MK.IV.BPM2.4.NB.4.BEHS.0093

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0295
SOM.MK.I.BPM2.4.NB.1.NSCI.0296
SOM.MK.I.BPM2.4.NB.1.NSCI.0297
SOM.MK.I.BPM2.4.NB.1.NSCI.0298
SOM.MK.I.BPM2.4.NB.1.NSCI.0299
SOM.MK.I.BPM2.4.NB.1.NSCI.0300

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0302
SOM.CS.I.BPM2.4.NB.1.NSCI.0303
SOM.MK.I.BPM2.4.NB.1.NSCI.0304
SOM.MK.I.BPM2.4.NB.1.NSCI.0305
SOM.MK.I.BPM2.4.NB.1.NSCI.0306
SOM.MK.IV.BPM2.4.NB.1.NSCI.0307

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0094

Trauma- and Stressor-Related Disorders & Dissociative Disorders

Apply the essential diagnostic criteria for the trauma- and stressor-related disorders, including any subtypes

Clinically differentiate the trauma- and stressor-related disorders (including any subtypes) from other conditions

Describe the biopsychosocial factors contributing to trauma- and stressor-related disorders

Describe the management of trauma- and stressor-related disorders, including mechanisms of action and potential side effects

Apply the essential diagnostic criteria for the dissociative disorders, including any subtypes

Clinically differentiate the dissociative disorders, including any subtypes from other conditions

Describe the biopsychosocial factors contributing to dissociative disorders

Describe the management for dissociative disorders, including mechanisms of action and potential side effects

Depressive Disorders & Bipolar and Related Disorders

Apply the essential diagnostic criteria for the depressive disorders

Clinically differentiate the depressive disorders from other conditions

Describe the major neuropathological changes in depression

Describe the management of depression, including mechanisms of action and potential side effects

Apply the essential diagnostic criteria for the bipolar and related disorders

Clinically differentiate the bipolar and related disorders from other conditions

Describe the biopsychosocial factors contributing to bipolar and related disorders

Describe the management of bipolar and related disorders, including mechanisms of action and potential side effects

Other Depressive Disorders, Specifiers for Mood Disorders; and Brain Stimulation Therapies for Mood Disorders

Apply the essential diagnostic criteria for the depressive disorders

Clinically differentiate the depressive disorders from other conditions

Apply the diagnostic specifiers for mood disorders, including melancholic, atypical, psychotic (mood-congruent and mood incongruent), catatonic, peripartum onset, seasonal pattern, and rapid cycling

Clinically differentiate specifiers for mood disorders, including melancholic, atypical, psychotic (mood-congruent and mood incongruent), catatonic, peripartum onset, seasonal pattern, and rapid cycling

Describe brain stimulation therapies for mood disorders, including electroconvulsive therapy, repetitive transcranial magnetic stimulation, and vagus nerve stimulation, and their potential side effects

Biological Rhythms

Describe two methods the brain uses to produce synchronized, rhythmic oscillations

Define ultradian, circadian and infradian rhythms

Describe input to and outputs from pacemaker systems

Discuss modulation of respiratory rhythmicity

Describe the physiological parameters that follow a circadian pattern

Discuss the role of the suprachiasmatic nucleus serves as a biological clock

Sleep

Describe hypotheses regarding the role of sleep

List techniques used in the assessment of brain activity during sleep

Contast sleep stages through the use of electroencephalography

Contrast neural mechanisms that support sleeping and waking states

Describe age-related changes to quality and quantity of sleep

Discuss the effects of sleep deprivation on daily activities

Sleep-Wake Disorders

Apply the essential diagnostic criteria for sleep-wake disorders

SOM.MK.I.BPM2.4.NB.4.BEHS.0182
SOM.MK.III.BPM2.4.NB.2.BEHS.0095
SOM.MK.IV.BPM2.4.NB.4.BEHS.0096

Clinically differentiate the sleep-wake disorders based on symptomatology and results of polysomnography
Describe the biopsychosocial factors contributing to sleep-wake disorders
Describe the management of sleep-wake disorders, including mechanisms of action and potential side effects

Small group BEHS
SOM.CS.III.BPM2.5.NB.5.BEHS.0165
SOM.CS.I.BPM2.5.NB.5.BEHS.0166
SOM.CS.III.BPM2.5.NB.5.BEHS.0167

NB SG 12 - Cognitive Screening Using the Mini–Mental State Examination (MMSE)
Demonstrate professional communication skills while interviewing and gathering information from a patient presenting with a cognitive concern
Introduce the formal cognitive testing component of the mental status examination and provide appropriate closing remarks at the end of the encounter
Administer, score, and interpret the Folstein Mini-Mental State Examination

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0308
SOM.MK.I.BPM2.4.NB.1.NSCI.0309
SOM.MK.I.BPM2.4.NB.1.NSCI.0310
SOM.MK.I.BPM2.4.NB.1.NSCI.0311
SOM.MK.II.BPM2.4.NB.1.NSCI.0312
SOM.CS.I.BPM2.4.NB.1.NSCI.0313
SOM.CS.I.BPM2.4.NB.1.NSCI.0314

Consciousness
Define consciousness and differentiate coma from other states of reduced consciousness
Describe the ascending reticular activating system
Discuss the roles of brainstem and thalamus in regulating consciousness and wake-sleep cycles.
Differentiate coma from brain death, locked-in-syndrome, and vegetative state
List causes of coma
Demonstrate the use of the Glasgow Coma Scale
Describe preliminary examination of the unconscious patient

DLA NSCI
SOM.CS.I.BPM2.4.NB.1.NSCI.0315

Examination of an unconscious patient 1 and 2
Describe thorough examination of the unconscious patient

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0424
SOM.CS.III.BPM2.4.NB.4.NSCI.0425
SOM.CS.II.BPM2.4.NB.4.NSCI.0426
SOM.CS.II.BPM2.4.NB.4.NSCI.0427
SOM.CS.II.BPM2.4.NB.4.NSCI.0428
SOM.CS.II.BPM2.4.NB.4.NSCI.0429

Intracranial Pressure
Critically appraise the symptoms and underlying pathology(ies) of increased intracranial pressure.
In the context of elevated intracranial pressure, describe examination of an unconscious patient.
In the context of elevated intracranial pressure, discuss extraocular function
In the context of elevated intracranial pressure, the describe ocular motor nuclei and other control units of extraocular movements in the brain stem and in the cerebral cortex.
In the context of elevated intracranial pressure, describe the function and characteristic parameters of saccadic eye movements and lateral gaze
In the context of elevated intracranial pressure, describe upper and lower motor neuron signs.

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0097
SOM.MK.II.BPM2.4.NB.3.BEHS.0100
SOM.MK.I.BPM2.4.NB.4.BEHS.0183
SOM.MK.III.BPM2.4.NB.3.BEHS.0098
SOM.MK.III.BPM2.4.NB.2.BEHS.0099
SOM.MK.IV.BPM2.4.NB.4.BEHS.0101

Feeding and Eating Disorders
Apply the essential diagnostic criteria for the feeding and eating disorders, including any subtypes
Identify the clinical features of a patient with an eating disorder
Clinically differentiate the feeding and eating disorders, including any subtypes
Describe the epidemiology, onset, course, and outcome of the feeding and eating disorders
Describe the biopsychosocial factors contributing to the feeding and eating disorders
Describe the management of feeding and eating disorders, including mechanisms of action and potential side effects

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0581
SOM.MK.I.BPM2.4.NB.1.NSCI.0582
SOM.MK.I.BPM2.4.NB.1.NSCI.0583
SOM.MK.I.BPM2.4.NB.1.NSCI.0584
SOM.MK.I.BPM2.4.NB.1.NSCI.0585
SOM.MK.I.BPM2.4.NB.1.NSCI.0586
SOM.MK.I.BPM2.4.NB.1.NSCI.0587

Sensory-Motor Integration
Integrate neuroanatomical knowledge of the corticospinal, corticobulbar, dorsal column-medial lemniscus, and anterolateral system projection tracts.
Compare and contrast anterior cerebral, middle cerebral and posterior cerebral artery lesions
Analyze the similarities and differences between medial and lateral pontine and medullary vascular lesions.
Categorize saccadic, vestibulo-ocular, and optokinetic eye movements as voluntary or reflex, describe their neuroanatomical pathways, and describe their assessment in the neurological examination.
Explain the phases and testing of physiological nystagmus
List four causes of pathological nystagmus and describe expected related observations during neurological examination
Apply knowledge of the corticospinal, corticobulbar, dorsal column-medial lemniscus, and anterolateral system tracts to localize a lesion in the central nervous system

SOM.MK.I.BPM2.4.NB.1.NSCI.0588	Apply knowledge of medial and lateral pontine and medullary vascular lesions to localize a lesion in the central nervous system
SOM.MK.I.BPM2.4.NB.1.NSCI.0589	Apply knowledge of saccadic, vestibulo-ocular, and optokinetic eye movements, and physiological and pathological nystagmus to localize a lesion in the central nervous system
Lecture BEHS	Psychotic Disorders I: Schizophrenia
SOM.MK.I.BPM2.4.NB.4.BEHS.0110	Describe the 5 psychotic symptom domains
SOM.MK.I.BPM2.4.NB.2.BEHS.0176	Describe the biopsychosocial factors contributing to psychotic disorders
SOM.MK.I.BPM2.4.NB.4.BEHS.0184	Clinically distinguish disorganized speech of psychosis from language disturbances associated with neurological disorders
SOM.MK.I.BPM2.4.NB.4.BEHS.0185	Apply the essential diagnostic criteria for schizophrenia
SOM.MK.I.BPM2.4.NB.4.BEHS.0115	Clinically differentiate the psychotic disorders from other conditions
SOM.MK.II.BPM2.4.NB.2.BEHS.0111	Describe the dopamine hypothesis of schizophrenia
SOM.MK.II.BPM2.4.NB.2.BEHS.0112	Discuss the common neuropathological findings in schizophrenia
Small group NSCI	NB SG 13 - Mini OSPE
SOM.CS.IV.BPM2.5.NB.5.NSCI.0580	Demonstrate the appropriate neurologic examination during a standardized patient encounter
Lecture BEHS	Psychotic Disorders II: Other Psychotic Disorders and Management
SOM.MK.IV.BPM2.4.NB.4.BEHS.0114	Describe the management of schizophrenia and other psychotic disorders, including mechanisms of action and potential side effects
SOM.MK.I.BPM2.4.NB.4.BEHS.0115	Clinically differentiate the psychotic disorders from other conditions
Lecture BEHS	Personality Disorders I: (Clusters A and B)
SOM.MK.I.BPM2.4.NB.4.BEHS.0121	Apply the essential diagnostic criteria for the Cluster A, Cluster B and Cluster C personality disorders
SOM.MK.III.BPM2.4.NB.2.BEHS.0196	Describe the biopsychosocial factors contributing to personality disorders
SOM.MK.IV.BPM2.4.NB.4.BEHS.0120	Describe the management of personality disorders, including mechanisms of action and potential side effects
SOM.MK.I.BPM2.4.NB.4.BEHS.0122	Clinically differentiate the personality disorders from other conditions
Lecture BEHS	Personality Disorders II: (Cluster C) & Disruptive, Impulse-Control, and Conduct Disorders
SOM.MK.I.BPM2.4.NB.4.BEHS.0121	Apply the essential diagnostic criteria for the Cluster A, Cluster B and Cluster C personality disorders
SOM.MK.I.BPM2.4.NB.4.BEHS.0122	Clinically differentiate the personality disorders from other conditions
SOM.CS.III.BPM2.4.NB.3.BEHS.0123	Describe the role of psychological tests in assessing personality traits
SOM.MK.I.BPM2.4.NB.4.BEHS.0196	Apply the essential diagnostic criteria for disruptive, impulse-control, and conduct disorders
SOM.MK.I.BPM2.4.NB.4.BEHS.0124	Clinically differentiate the disruptive, impulse-control, and conduct disorders from other conditions, including disruptive mood dysregulation disorder and adjustment disorder
SOM.MK.III.BPM2.4.NB.2.BEHS.0125	Describe the biopsychosocial factors contributing to oppositional and conduct-disturbed behavior
SOM.MK.IV.BPM2.4.NB.4.BEHS.0126	Describe the management of oppositional and conduct-disturbed behavior, including mechanisms of action and potential side effects
Lecture BEHS	Neurodevelopmental Disorders
SOM.MK.I.BPM2.4.NB.4.BEHS.0130	Apply the essential diagnostic criteria to intellectual disability (intellectual developmental disorder) and global developmental delay
SOM.MK.I.BPM2.4.NB.4.BEHS.0191	Clinically differentiate intellectual disability (intellectual developmental disorder) and global developmental delay
SOM.MK.I.BPM2.4.NB.4.BEHS.0131	Apply the essential diagnostic criteria to language disorder, speech-sound disorder, social (pragmatic) communication disorder, and childhood-onset fluency disorder
SOM.MK.I.BPM2.4.NB.4.BEHS.0192	Clinically differentiate language disorder, speech-sound disorder, social (pragmatic) communication disorder, and childhood-onset fluency disorder
SOM.MK.I.BPM2.4.NB.4.BEHS.0132	Apply the essential diagnostic criteria and specifiers to autism spectrum disorder
SOM.MK.I.BPM2.4.NB.4.BEHS.0133	Apply the essential diagnostic criteria and specifiers to specific learning disorder
SOM.MK.I.BPM2.4.NB.4.BEHS.0134	Apply the essential diagnostic criteria and specifiers to attention-deficit/hyperactivity disorder (ADHD)
SOM.MK.IV.BPM2.4.NB.4.BEHS.0135	Describe the management for ADHD, including mechanisms of action and potential side effects
SOM.MK.I.BPM2.4.NB.4.BEHS.0171	Apply the essential diagnostic criteria for Tourette's disorder
SOM.MK.IV.BPM2.4.NB.4.BEHS.0172	Describe the management of Tourette's disorder, including mechanisms of action and potential side effects

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0321
SOM.MK.I.BPM2.4.NB.1.NSCI.0322
SOM.MK.I.BPM2.4.NB.1.NSCI.0323
SOM.MK.I.BPM2.4.NB.1.NSCI.0324
SOM.MK.I.BPM2.4.NB.1.NSCI.0325
SOM.MK.I.BPM2.4.NB.1.NSCI.0326
SOM.MK.I.BPM2.4.NB.1.NSCI.0327

Lecture BEHS
SOM.MK.I.BPM2.4.NB.4.BEHS.0136
SOM.MK.I.BPM2.4.NB.4.BEHS.0193
SOM.MK.II.BPM2.4.NB.2.BEHS.0137
SOM.MK.II.BPM2.4.NB.2.BEHS.0194
SOM.MK.IV.BPM2.4.NB.4.BEHS.0138
SOM.MK.I.BPM2.4.NB.4.BEHS.0139
SOM.MK.II.BPM2.4.NB.2.BEHS.0140
SOM.MK.IV.BPM2.4.NB.4.BEHS.0141

Small group BEHS
SOM.CS.III.BPM2.4.NB.5.BEHS.0042
SOM.CS.II.BPM2.4.NB.4.BEHS.0168
SOM.CS.II.BPM2.4.NB.4.BEHS.0169

Lecture NSCI
SOM.MK.I.BPM2.4.NB.1.NSCI.0328
SOM.MK.I.BPM2.4.NB.1.NSCI.0329
SOM.MK.I.BPM2.4.NB.1.NSCI.0330
SOM.MK.I.BPM2.4.NB.1.NSCI.0331

DLA NSCI
SOM.CS.II.BPM2.4.NB.4.NSCI.0449
SOM.CS.II.BPM2.4.NB.4.NSCI.0450
SOM.CS.II.BPM2.4.NB.4.NSCI.0451
SOM.CS.III.BPM2.4.NB.4.NSCI.0452
SOM.CS.II.BPM2.4.NB.4.NSCI.0453
SOM.CS.II.BPM2.4.NB.4.NSCI.0454
SOM.CS.II.BPM2.4.NB.4.NSCI.0455

Lecture BEHS

SOM.MK.I.BPM2.4.NB.4.BEHS.0142
SOM.MK.II.BPM2.4.NB.2.BEHS.0173
SOM.MK.IV.BPM2.4.NB.4.BEHS.0143

Memory/Amnesia

Distinguish declarative from non-declarative memory and working memory from short-term and long-term memory
Relate damage in diencephalic and hippocampal systems to different amnesia profiles
Define the concept ‘engram’ and explain how and where engrams are stored in the brain
Explain the neural connections in hippocampus that are believed to be involved in memory generation
Discuss the cellular/molecular basis of long-term potentiation in the hippocampus
Discuss the cellular/molecular basis of long-term depression in the hippocampus
Explain the role of [CA2+] long-term potentiation and long-term depression

Neurocognitive Disorders I: Delirium and Alzheimer’s Dementia

Apply the essential diagnostic criteria for delirium
Clinically differentiate delirium and dementia
Discuss common causes of a delirium
Identify the underlying neuropathology in delirium
Describe the management of delirium, including mechanisms of action and potential side effects
Describe the clinical features of Alzheimer’s dementia, including the typical progression of symptoms through early, middle, and late stages of the disorder
Describe neuropathological changes in Alzheimer’s dementia
Describe the management of Alzheimer’s dementia, including mechanisms of action and potential side effects

NB SG 14 - Psychopathology Diagnosis

Apply terminology for abnormalities in mental status domains (e.g., appearance, behavior, mood/affect, thought, perception, insight, judgment, cognition) to clinical scenarios
Identify the most likely psychiatric diagnosis for a patient based on clinical presentation
Explain how major differential diagnoses of psychiatric disorders are ruled out based on a patient’s clinical presentation

Language/Aphasia

Contrast the roles of the left and right hemisphere in processing language
Describe the neuroanatomical networks underlying speaking, hearing, reading, writing
Contrast the aphasias based of locations of lesions
Describe the neuroanatomical regions implicated in reading disorders

Wernicleke's Encephalopathy

Critically appraise the symptoms and underlying pathology(ies) of thiamine deficiency.
In the context of Wernicke's encephalopathy, compare and contrast disorders associated with brain tissue: Alzheimer disease, Prion diseases, multiple sclerosis, Wernicke
In the context of Wernicke's encephalopathy, describe the epidemiology, onset, progression, associated features and predictors of outcome.
In the context of Wernicke's encephalopathy, apply the diagnostic criteria for a delirium.
Apply knowledge of nystagmus to Wernicke's encephalopathy.
In the context of Wernicke's encephalopathy, explain chronic forms of neuropathy (e.g., Hansen disease, lead poisoning, alcoholic, diabetes mellitus).
In the context of Wernicke's encephalopathy, correlate cerebellar lesion sites and clinical manifestations.

Neurocognitive Disorders II: Other Types of Dementia

Differentiate Alzheimer’s dementia from vascular, frontotemporal, Lewy body, Parkinson’s, Huntington’s, and prion-related dementia, based on predominant clinical presentation, laboratory studies, and imaging studies
Describe neuropathological changes in vascular, frontotemporal, Lewy body, Parkinson’s, Huntington’s, and prion-related dementia
Describe the management of vascular, frontotemporal, Lewy body, Parkinson’s, Huntington’s, and prion-related dementia, including mechanisms of action and potential side effects

SOM.MK.I.BPM2.4.NB.4.BEHS.0144 Differentiate dementia from normal age-related cognitive changes, mild cognitive impairment, and depression-related cognitive impairment
SOM.MK.I.BPM2.4.NB.4.BEHS.0145 Differentiate the diagnosis of neurocognitive disorder (including major and mild subtypes) from dementia
SOM.MK.I.BPM2.4.NB.2.BEHS.0146 Identify the relationship between brain structures and their associated neuropsychological function

Lecture BEHS

SOM.MK.I.BPM2.4.NB.4.BEHS.0147
SOM.MK.III.BPM2.4.NB.2.BEHS.0148
SOM.MK.I.BPM2.4.NB.4.BEHS.0149
SOM.MK.IV.BPM2.4.NB.4.BEHS.0150
SOM.MK.I.BPM2.4.NB.4.BEHS.0151
SOM.MK.I.BPM2.4.NB.4.BEHS.0152
SOM.MK.I.BPM2.4.NB.4.BEHS.0153
SOM.MK.IV.BPM2.4.NB.4.BEHS.0174

Substance-Related and Addictive Disorders: Depressants and Stimulants

Differentiate the terms misuse, intoxication, withdrawal, substance use disorder, and dependence (psychological and physical dependence)

Describe the biopsychosocial factors contributing to substance-use disorders

Differentiate between symptoms of intoxication and withdrawal for depressant drugs (alcohol, benzodiazepines, barbiturates) and inhalants

Identify the medications used to treat alcohol use disorder, alcohol withdrawal, and benzodiazepine toxicity, including mechanisms of action and potential side effects

Clinically differentiate between intoxication and withdrawal with major psychostimulant drugs (e.g., cocaine, methamphetamine)

Describe the effects of MDMA (Ecstasy) and other contemporary drugs of abuse, such as bath salts and the designer drug K2

Clinically differentiate between intoxication and withdrawal with minor psychostimulant drugs (caffeine, nicotine)

Identify the medications used to treat nicotine use disorder, including mechanisms of action and potential side effects

Lecture BEHS

SOM.MK.III.BPM2.4.NB.3.BEHS.0154
SOM.MK.I.BPM2.4.NB.4.BEHS.0155
SOM.MK.IV.BPM2.4.NB.4.BEHS.0175
SOM.MK.I.BPM2.4.NB.4.BEHS.0156
SOM.MK.IV.BPM2.4.NB.4.BEHS.0157
SOM.CS.III.BPM2.4.NB.3.BEHS.0158
SOM.MK.IV.BPM2.4.NB.4.BEHS.0159
SOM.MK.I.BPM2.4.NB.4.BEHS.0160
SOM.MK.IV.BPM2.4.NB.4.BEHS.0161

Substance-Related and Addictive Disorders: Hallucinogens, Dissociative Anesthetics, Cannabis, Opioids, Gambling, and Substance-Use Management

Summarize the schedules of controlled substances under the Controlled Substances Act and provide examples of each, including the status of medicinal cannabis

Clinically differentiate between intoxication and withdrawal with hallucinogens, cannabis, and dissociative anesthetics

Describe the management of hallucinogen-related and similar disorders, including mechanisms of action and potential side effects

Clinically differentiate between intoxication and withdrawal with opioid narcotics

Identify the medications used to treat acute opioid overdose and opioid use disorder, including mechanisms of action and potential side effects

Evaluate the safety of a patient’s alcohol and drug use based on results of screening tools

Identify management approaches used in drug rehabilitation

Apply the essential diagnostic criteria for gambling disorder

Describe the management of gambling disorder, including mechanisms of action and potential side effects